A Reduced-Rank Panel Data Model with Interactive Fixed Effects

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Abstract

This paper studies a panel data model with interactive fixed effects and heterogeneous slopes. To balance the flexibility of heterogeneous coefficient models and the efficiency of homogeneous specifications, we propose a Reduced-Rank Panel Model with Interactive Fixed Effects (RRPM-IFEs), in which individual-specific slope coefficients are driven by low-dimensional linear combinations of the covariates. This reduced-rank structure captures structured cross-sectional heterogeneity while achieving dimension reduction, and nests homogeneous and fully heterogeneous panel models as special cases.

We study the model identification and propose a Constrained Principal Component (CPC) estimator for estimation. We derive asymptotics of the proposed estimator and develop feasible inference procedures with bias correction. We further propose model selection criteria for determining the structural dimensions of slope heterogeneity and interactive fixed effects. Simulation studies demonstrate that the proposed method delivers effective dimension selection, and empirical applications illustrate how the reduced-rank structure uncovers interpretable latent patterns in heterogeneous slopes.

Keywords: Bias correction; Interactive fixed effects; Model selection; Panel data; Reduced-rank structure; Slope heterogeneity.

1 Introduction

Panel data models play a central role in empirical analysis by exploiting both cross-sectional and temporal variation, thereby allowing one to account for unobserved heterogeneity and improve estimation efficiency. Panel models with Interactive Fixed Effects (IFEs) extend this framework by introducing latent common factors with heterogeneous loadings, providing a flexible approach to modeling cross-sectional dependence and unobserved global shocks; see, for example, Pesaran (2006), Bai (2009), and Moon and Weidner (2015).

Suppose y_{it} and $\boldsymbol{x}_{it}$ are the response of interest and covariates, with $\dim(\boldsymbol{x}_{it}) = p$ observed from the i–th subject at time t, where $i = 1, \ldots, N$, $t = 1, \ldots, T$. A general variable-coefficient panel data model with IFEs is given by

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\beta}_{it,0} + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \varepsilon_{it}, \quad i = 1, \ldots, N, \, t = 1, \ldots, T, \tag{1.1}$$

where the subscript "0" denotes the true parameters values, $\boldsymbol{\beta}_{it,0}$ is a p-dimensional coefficient vector for section i at time t, ε_{it} is the idiosyncratic error, and $\boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0}$ constitutes IFEs with both $\boldsymbol{l}_{i,0}$ and $\boldsymbol{f}_{t,0}$ being

R_0-dimensional vectors. Since the unobservable latent factors at time t, $\boldsymbol{f}_{t,0}$, affects each individual with different intensities, that is, the unobservable factor loadings $\boldsymbol{l}_{i,0}$, this factor structure is also interpreted as time-varying individual effects.

To ensure that the model is identifiable and estimable, certain restrictions must be imposed on the coefficients $\{\boldsymbol{\beta}_{it,0}\}_{i,t}$; see Chapter 13 in Hsiao (2022) for an overview. The first class of methods allows coefficients to vary over time. One approach assumes that $\boldsymbol{\beta}_{it,0}$ is a smooth function of $\tau_t := t/T$, that is, $\boldsymbol{\beta}_{it,0} = \boldsymbol{\beta}_0(\tau_t)$, leading to time-varying coefficient panels; see, e.g., Liu et al. (2020) and Wang et al. (2025). An alternative allows $\boldsymbol{\beta}_{it,0} = \boldsymbol{\beta}_{t,0}$ to experience structural breaks over time; see, e.g., Li et al. (2016), Boldea et al. (2020), and Karavias et al. (2023). The second class of methods assumes that the coefficients vary only across sections, that is, $\boldsymbol{\beta}_{it,0} = \boldsymbol{\beta}_{i,0}$. Under this restriction, model (1.1) reduces to

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\beta}_{i,0} + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \varepsilon_{it}, \quad i = 1, \ldots, N,\ t = 1, \ldots, T. \tag{1.2}$$

See, e.g., Song (2013), Ando and Bai (2015), and Avarucci and Zaffaroni (2019) for analyses of panel models with fully heterogeneous coefficients across sections and IFEs. Other structures have also been considered. For instance, Chernozhukov et al. (2019) impose factor structures, analogues to IFEs, on the coefficients, leading to the model:

$$y_{it} = \sum_{j=1}^{p} \left[x_{it,j} \cdot \left(\boldsymbol{\alpha}_{ij,0}^\top \boldsymbol{\pi}_{tj,0} \right) \right] + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \varepsilon_{it}, \quad i = 1, \ldots, N,\ t = 1, \ldots, T, \tag{1.3}$$

where $\{\boldsymbol{\alpha}_{ij,0}\}_{i=1}^N$ and $\{\boldsymbol{\pi}_{tj,0}\}_{i=1}^N$ are loading-factor pairs for each predictor j. Dong et al. (2021) assumes that $\boldsymbol{\beta}_{it,0}$ is an unknown function of observable r_{it}. Both settings allow the coefficients to vary across (i, t).

1.1 Motivations and Relations to Existing Models

Our discussion focuses on the case where the coefficients vary across sections. Traditional panel models further assume common slopes are shared across sections, that is, $\boldsymbol{\beta}_{i,0} = \boldsymbol{\beta}_0$, leading to the model

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\beta}_0 + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \varepsilon_{it}, \quad i = 1, \ldots, N,\ t = 1, \ldots, T. \tag{1.4}$$

Although estimators of common slopes are $\sqrt{NT}$-consistent in theory (Bai, 2009; Bai and Li, 2014; Moon and Weidner, 2017), empirical studies find evidence against slope homogeneity, showing that neglecting coefficient heterogeneity may lead to inconsistent estimators, unreliable inference, and unconvincing conclusions. For instance, Robertson and Symons (1992) analyze a basic real wage equation using a panel of countries and find that imposing slope homogeneity introduces either upward or downward bias in the parameter estimates. Hsiao and Tahmiscioglu (1997) examine various forms of the investment function and reject the slope homogeneity hypothesis across firms. Durlauf et al. (2001) demonstrate that empirical analysis of the Solow

growth model supports accounting for cross-country parameter heterogeneity, favoring a local generalization over the conventional global specification with homogeneous parameters. Similarly, Su and Chen (2013) examine an economic growth model with a latent factor structure $\boldsymbol{l}_{i,0}^{\top}\boldsymbol{f}_{t,0}$, and reject the null hypothesis of slope homogeneity based on a Lagrange Multiplier test.

While allowing for complete slope heterogeneity in (1.2) avoids model misspecification by capturing individual-level differences, it often leads to imprecise estimates due to short panels. Moreover, ignoring the pooling of information across units further reduces statistical efficiency and makes it harder to uncover common patterns in panels. These issues are well documented: Browning and Carro (2007) caution that allowing heterogeneity requires balancing data fit and interpretability; Su et al. (2016) and Wang et al. (2018) emphasize that permitting complete cross-sectional slope heterogeneity sacrifices the benefits of cross-sectional averaging and can make estimation impractical with a limited time dimension; Sarafidis and Weber (2015) note that assuming either complete heterogeneity or homogeneity may be unsatisfactory in some applications; see, e.g., Burnside (1996) and Baltagi and Griffin (1997).

However, the choice between slope homogeneity and complete heterogeneity in slopes need not be an either-or decision. One intermediate approach is the random coefficient model, in which individual-specific slopes are random but share a common expectation, that is, $\mathbb{E}(\boldsymbol{\beta}_{1,0}) = \cdots = \mathbb{E}(\boldsymbol{\beta}_{N,0})$; see, e.g., Swamy (1970), Chamberlain (1992), and Pesaran (2006). Alternatively, panel data models with structured heterogeneity can be employed. A widely studied example is the latent group structure, where coefficients are homogeneous within groups but heterogeneous across them; see, e.g., Lin and Ng (2012), Su and Ju (2018), and Huang et al. (2021) for related studies.

In this paper, we propose a reduced-rank structure for $\{\boldsymbol{\beta}_{i,0}\}_{i=1}^{N}$ to resolve the fundamental bias-efficiency trade-off in panel modeling: homogeneous models cause specification bias when slopes truly vary, while fully heterogeneous models yield inefficient estimates. Specifically, we assume

$$\boldsymbol{\beta}_{i,0} = \sum_{k=1}^{K_0} \boldsymbol{\theta}_{k,0} b_{ik,0}, \quad K_0 \le p,$$

where K_0 is the true rank of the heterogeneous coefficient structure, $\boldsymbol{\theta}_{k,0} \in \mathbb{R}^p$ represents common characteristics, and $b_{ik,0}$ accounts for individual heterogeneity. This formulation permits heterogeneous responses across units while maintaining a parsimonious low-dimensional representation. Based on this structure, we introduce the Reduced-Rank Panel Model (RRPM):

$$y_{it} = \sum_{k=1}^{K_0} \boldsymbol{x}_{it}^{\top}\boldsymbol{\theta}_{k,0} b_{ik,0} + \varepsilon_{it}, \quad i = 1, \ldots, N,\ t = 1, \ldots, T, \tag{1.5}$$

where $\boldsymbol{\theta}_{k,0}$, $k = 1, \ldots, K_0$ are the global common projections that cross over all the individuals, but the effect on each individual, $b_{ik,0}$, varies.

In terms of model form, RRPM coincides with the Seemingly UnRElated Reduced-Rank regression (SURERR) model of Velu and Richards (2008), which can be viewed as Reduced-Rank Regression (RRR; Izenman, 1975) applied in a Seemingly Unrelated Regression (SUR; Zellner, 1962) framework. Let $\boldsymbol{X}_i = \left(\boldsymbol{x}_{i1}, \quad \ldots, \quad \boldsymbol{x}_{iT}\right)^\top$. When $\boldsymbol{X}_1 = \cdots = \boldsymbol{X}_N$, RRPM reduces to classical RRR. While RRPM allows for cross-sectional variation in predictors, it retains the key advantages of RRR due to the reduced-rank structure, such as parameter parsimony and the ability to capture latent structure. Empirical support for the reduced-rank structure is provided by Velu and Richards (2008), who regress weekly sales on multiple predictors and select a rank-one model based on information criteria, uncovering a dominant market factor driven mainly by price variables. Although Reinsel et al. (2022) note that SURERR or RRPM is broadly applicable and naturally suited to panel settings, research on RRPM remains limited and warrants further investigation. A similar idea underlying RRPM also appears in the framework of Multi-Task Feature Learning (MTFL). In Argyriou et al. (2008), each cross-sectional unit is interpreted as a separate task, represented by a function $f_i : \mathbb{R}^p \to \mathbb{R}$, with

$$f_i\left(\boldsymbol{x}\right) = \sum_{k=1}^{K_0} b_{ik,0} h_k\left(\boldsymbol{x}\right), \quad i = 1, \ldots, N,$$

where $\{h_k : \mathbb{R}^p \to \mathbb{R}\}_{k=1}^{K_0}$ are the shared features, and $\{b_{ik,0}\}_{k=1}^{K_0}$ are task-specific coefficients. In particular, they consider the case where the features $h_k(\boldsymbol{x})$ are linear function, that is, $h_k\left(\boldsymbol{x}\right) = \boldsymbol{x}^\top \boldsymbol{\theta}_{k,0}$, which closely mirrors the reduced-rank structure.

Combining (1.5) with the multi-factor structure, $\boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0}$, leads to the following Reduced-Rank Panel Model with Interactive Fixed Effects (RRPM-IFEs):

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \varepsilon_{it}, \quad i = 1, \ldots, N, \, t = 1, \ldots, T. \tag{1.6}$$

Here $\boldsymbol{\Theta}_0^\top \boldsymbol{x}_{it}$ are some linear combinations of observable predictors $\boldsymbol{x}_{it}$, whereas $\boldsymbol{f}_{t,0}$ is completely unobservable; thus the two should be strictly distinguished, even though both function as factors in a broad sense. RRPM-IFEs nests important benchmark models. When $K_0 = 1$ and $b_{i1,0} = 1$ for all i, the model collapses to the homogeneous panel model with IFEs in (1.4). When $K_0 = p$, it reduces to the fully heterogeneous panel model with IFEs in (1.2). Hence, RRPM-IFEs provides a flexible framework that spans the full spectrum between homogeneity and complete heterogeneity.

1.2 Contributions

In panel models with IFEs, ignoring latent factors generally leads to inconsistency when the predictors correlate with $\boldsymbol{f}_{t,0}$ or $\boldsymbol{l}_{i,0}$. Consequently, numerous methods have been developed to account for such latent structures. For panels with large T, two widely used approaches are Principal Components (PC) estimators

(Bai, 2009; Su and Chen, 2013; Moon and Weidner, 2015) and Common Correlated Effects (CCE) estimators (Pesaran, 2006; Chudik et al., 2011; De Vos and Everaert, 2021). For panels with fixed T, many estimators rely on the Generalized Method of Moments (GMM); see Juodis and Sarafidis (2018) for an overview. In this paper, we adopt the PC approach because of its transparent algorithmic structure, ease of implementation, and appealing theoretical properties.

The incidental parameters problem (IPP; Neyman and Scott, 1948) arises when structural parameters are estimated jointly with high-dimensional nuisance parameters, such as individual and time effects in panel models, leading to non-negligible bias in the structural estimates. In dynamic or weakly exogenous specifications, an additional Nickell bias (Nickell, 1981) further contaminates estimation. Under rectangle-array asymptotics $(N, T) \to \infty$ with $T/N \to \rho \in (0, \infty)$, both biases persist and are of the same order as the standard deviation, rendering bias correction essential for valid inference.

While existing bias correction methods address special cases, such as models without IFEs (Hahn and Newey, 2004; Hahn and Kuersteiner, 2011) or with homogeneous slopes (Bai, 2009; Moon and Weidner, 2017), general results for settings combining interactive effects and structured coefficient heterogeneity remain limited. In particular, bias correction developed by Fernández-Val and Weidner (2016) and Chen et al. (2021) cannot be directly applied due to fundamental structural differences. Although Sun (2016) and Jochmans and Otsu (2019) propose likelihood-corrected estimators for two-way fixed effects, formal theoretical results are still unavailable.

A large literature has studied the determination of the number of latent factors R_0 in pure (approximate) factor models: $y_{it} = \boldsymbol{l}_{i,0}^{\top} \boldsymbol{f}_{t,0} + \varepsilon_{it}$; see, e.g., Bai and Ng (2002), Onatski (2010), and Ahn and Horenstein (2013). Here "pure" refers to models in which only one structural component is present, that is, $K_0 = 0$ or $R_0 = 0$. In panel models with IFEs, correct specification of R_0 is also crucial, as the performance of the PC estimator of Bai (2009) may be sensitive to factor misspecification; see, e.g., Lu and Su (2016). Aside from this line of work, the determination of the number of common components K_0 in pure RRPM has also been examined; see, e.g., Velu and Richards (2008) in the SURERR setting and Argyriou et al. (2008) in the MTFL setting.

However, the joint selection of (K, R) in models that simultaneously feature a reduced-rank coefficient structure and interactive effects has, to our knowledge, not been studied. The work closest to ours is Chernozhukov et al. (2019), who impose factor structures on regression coefficients in (1.3) and use nuclear norm regularization to estimate the corresponding low-rank matrices. Their primary focus is the inference on heterogeneous components, whereas our emphasis lies on selecting the structural ranks (K, R).

Our contributions are twofold. On the modeling side, we study a panel data model with IFEs and structured heterogeneous coefficients. Relative to existing approaches, our framework integrates several im-

portant features. By allowing both low-rank and high-rank covariates (see Section 2 and the `French` and `PWT` applications in Section 4.2), RRPM unifies RRR (Izenman, 1975) with panel data models in which each unit is associated with its own distinct observations, thus substantially enhancing modeling flexibility. Compared with the SURERR framework (Velu and Richards, 2008), our model explicitly incorporates IFEs and allows both $(N, T) \to \infty$. In contrast to the homogeneous slope model (Bai, 2009), we allow for slope heterogeneity across sections – a feature more reflective of empirical applications. Compared with the fully heterogeneous model (Song, 2013), our model imposes a reduced-rank structure on the heterogeneous coefficients, preserving meaningful cross-sectional patterns while achieving dimension reduction. Finally, relative to latent group models such as Su and Ju (2018), our reduced-rank framework provides an alternative approach to characterizing structured heterogeneity, offering a complementary perspective that is less explored in the presence of IFEs.

On the theoretical side, we establish the asymptotic normality of the proposed Constrained Principal Component (CPC) estimator under rectangle-array asymptotics, explicitly characterizing the incidental-parameter bias and the Nickell bias arising from dynamic specifications or violations of strict exogeneity. Building on this result, we derive an analytical bias correction for CPC estimation in settings featuring interactive fixed effects and reduced-rank coefficient heterogeneity. Moreover, we propose a class of model selection criteria (MSC), in the spirit of Bai and Ng (2002), for jointly selecting the pair (K, R) in the RRPM-IFEs, where K denotes the rank of the reduced-rank coefficient structure and R the number of latent factors.

1.3 Notation

For a symmetric positive definite matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\lambda_{\max}(\boldsymbol{A})$ and $\lambda_{\min}(\boldsymbol{A})$ are the largest and smallest eigenvalues of $\boldsymbol{A}$ respectively. For constants c_1, c_2, $c_1 \le \boldsymbol{A} \le c_2$ means that $\boldsymbol{A} - c_1\boldsymbol{I}_n$ and $c_2\boldsymbol{I}_n - \boldsymbol{A}$ are positive semi-definite, where $\boldsymbol{I}_n$ is an $n \times n$ identity matrix. For any matrix $\boldsymbol{A} \in \mathbb{R}^{m\times n}$, $\boldsymbol{A}^+$ denotes its Moore-Penrose inverse. $\|\boldsymbol{A}\|_2$ and $\|\boldsymbol{A}\|_f$ are spectral norm and Frobenius norm, respectively. For $1 \le l < m$, $\boldsymbol{A}_{[l]}$ denotes the first l rows of $\boldsymbol{A}$, while $\boldsymbol{A}_{-[l]}$ denotes the last $(m-l)$ rows of $\boldsymbol{A}$. $\mathcal{M}_{m,n}(\cdot)$ reshapes a vector of length mn into an $m \times n$ matrix, and $\ell(\cdot)$ vectorizes a matrix in turn. $\boldsymbol{P}_{\boldsymbol{A}}$ is the projection matrix onto the column space of $\boldsymbol{A}$, and $\boldsymbol{P}_{\boldsymbol{A}}^{\perp}$ is its orthogonal complement. $\boldsymbol{P}_{(\boldsymbol{A}_1,\boldsymbol{A}_2)}$ is the projection of the concatenated matrix $\begin{pmatrix}\boldsymbol{A}_1, & \boldsymbol{A}_2\end{pmatrix}$. The same notation applies to projections involving more matrices. $\boldsymbol{A}_3 \otimes \boldsymbol{A}_4$ is the Kronecker product of $\boldsymbol{A}_3$ and $\boldsymbol{A}_4$. We use M_{NT} and m_{NT} to denote $\max\left(\sqrt{N}, \sqrt{T}\right)$ and $\min\left(\sqrt{N}, \sqrt{T}\right)$, respectively. Let $[n] = \{1, 2, \ldots, n\}$, and $|\cdot|$ denote the cardinality of a set. $C_1, c_1, C_2, c_2, \ldots$ are generic positive constants that may differ in context. Throughout the paper, parameters with subscript "0" denote their true values.

The rest of the paper is organized as follows. Section 2 formally introduces the reduced-rank panel model

with interactive fixed effects. Section 3 presents the Constrained Principal Component (CPC) estimator and studies its theoretical properties. This section also proposes model selection criteria for jointly determining (K, R) when true values are unknown. Section 4 reports simulation results and empirical applications, and Section 5 concludes. Technical details are relegated to the supplementary material.

2 The Proposed Model and Its Identification

Before presenting the model, we first introduce some notation and definitions. Let x_{it} denote the j–th predictor of $\boldsymbol{x}_{it,j}$, and define the corresponding $T \times N$ matrix $\boldsymbol{X}_{\cdot j}$ as the collection of the j–th predictor across all individuals and time periods:

$$\boldsymbol{X}_{\cdot j} = \begin{pmatrix} \boldsymbol{x}_{1,j}, & \ldots, & \boldsymbol{x}_{N,j} \end{pmatrix} = \begin{pmatrix} x_{11,j} & \ldots & x_{N1,j} \\ \vdots & \ddots & \vdots \\ x_{1T,j} & \ldots & x_{NT,j} \end{pmatrix}.$$

Following Moon and Weidner (2017), we divide the p predictors into two types: p_1 high-rank predictors, and $p_2 = p - p_1$ low-rank predictors. For simplicity, we define $\boldsymbol{x}^{high} = (x_{\cdot 1}, \ \ldots, \ x_{\cdot p_1})^\top$ and $\boldsymbol{x}^{low} = (x_{\cdot p_1+1}, \ \ldots, \ x_{\cdot p})^\top$. As the name "low-rank" suggests, we consider predictors that admit a low-rank matrix representation. In the simplest and most common case, we assume that $\text{Rank}(\boldsymbol{X}_{\cdot j}) = 1$, so that $\boldsymbol{X}_{\cdot j} = \boldsymbol{g}_{\cdot j}\boldsymbol{\gamma}_{\cdot j}^\top$ with $\boldsymbol{g}_{\cdot j} \in \mathbb{R}^T$ and $\boldsymbol{\gamma}_{\cdot j} \in \mathbb{R}^N$, for $j = p_1 + 1, \ldots, p$. Two particularly common rank-one predictors are time-invariant predictors and cross-sectional invariant predictors, corresponding to $\boldsymbol{g}_{\cdot j}$ or $\boldsymbol{\gamma}_{\cdot j}$ being an all-one vector, respectively. Although Moon and Weidner (2017) argue that rank-one structure is sufficient in most applications, we later discuss how to accommodate more general low-rank predictors with finite rank; see Example 2.2. Let

$$\boldsymbol{\Gamma} = \begin{pmatrix} \boldsymbol{\gamma}_{\cdot p_1+1}, & \ldots, & \boldsymbol{\gamma}_{\cdot p} \end{pmatrix} = \begin{pmatrix} \boldsymbol{\gamma}_1, & \ldots, & \boldsymbol{\gamma}_N \end{pmatrix}^\top, \quad \boldsymbol{G} = \begin{pmatrix} \boldsymbol{g}_{\cdot p_1+1}, & \ldots, & \boldsymbol{g}_{\cdot p} \end{pmatrix} = \begin{pmatrix} \boldsymbol{g}_1, & \ldots, & \boldsymbol{g}_T \end{pmatrix}^\top, \tag{2.1}$$

For the high-rank predictors, $\boldsymbol{X}_{\cdot j}$ has a sufficiently high rank for $j = 1, \ldots, p_1$, so that Assumption 5 (2) is satisfied. This assumption would be violated if there exist a nonzero vector $\boldsymbol{\alpha} \in \mathbb{R}^{p_1}$ such that $\text{Rank}\left(\sum_{j=1}^{p_1} \alpha_j \boldsymbol{X}_{\cdot j}\right) \leq R_0$. This observation highlights a key distinction between the high-rank predictors and the low-rank predictors defined before. Let

$$\boldsymbol{X}_i^{high} = \begin{pmatrix} \boldsymbol{x}_{i,1}, & \ldots, & \boldsymbol{x}_{i,p_1} \end{pmatrix} \in \mathbb{R}^{T \times p_1}, \quad \boldsymbol{X}_i^{low} = \begin{pmatrix} \boldsymbol{x}_{i,p_1+1}, & \ldots, & \boldsymbol{x}_{i,p} \end{pmatrix} \in \mathbb{R}^{T \times p_2},$$
$$\boldsymbol{X}_i = \begin{pmatrix} \boldsymbol{X}_i^{high}, & \boldsymbol{X}_i^{low} \end{pmatrix} = \begin{pmatrix} \boldsymbol{x}_{i1}, & \ldots, & \boldsymbol{x}_{iT} \end{pmatrix}^\top, \quad \boldsymbol{y}_i = (y_{i1}, \ldots, y_{iT})^\top, \quad \boldsymbol{\varepsilon}_i = (\varepsilon_{i1}, \ldots, \varepsilon_{iT})^\top,$$

for each $i = 1, \ldots, N$, and $\boldsymbol{B}_0 = \begin{pmatrix} \boldsymbol{b}_{1,0}, & \ldots, & \boldsymbol{b}_{N,0} \end{pmatrix}^\top$, $\boldsymbol{F}_0 = \begin{pmatrix} \boldsymbol{f}_{1,0}, & \ldots, & \boldsymbol{f}_{T,0} \end{pmatrix}^\top$, $\boldsymbol{L}_0 = \begin{pmatrix} \boldsymbol{l}_{1,0}, & \ldots, & \boldsymbol{l}_{N,0} \end{pmatrix}^\top$. Then, (1.6) can be written as

$$\boldsymbol{y}_i = \boldsymbol{X}_i \boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} + \boldsymbol{F}_0 \boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i, \quad \text{s.t. } \boldsymbol{F}_0 \perp \boldsymbol{G}, \quad i = 1, \ldots, N, \tag{2.2}$$

where $\boldsymbol{F}_0 \perp \boldsymbol{G}$ indicates that $\boldsymbol{F}_0^\top \boldsymbol{G} = \boldsymbol{0}_{R_0 \times p_2}$. It is uncommon to impose an orthogonality condition between $(\boldsymbol{F},\ \boldsymbol{G})$ in panel models with IFEs, but such constraint is crucial for model identification when low-rank predictors are present. In panels with homogeneous slopes (1.4), identification can be ensured by imposing $N^{-1}\boldsymbol{L}_0^\top \boldsymbol{P}_{\boldsymbol{\Gamma}}^\perp \boldsymbol{L}_0 > 0$ and $T^{-1}\boldsymbol{F}_0^\top \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{F}_0 > 0$ for low-rank predictors (Moon and Weidner, 2017). However, in panels with heterogeneous slopes, low-rank predictors can cause identification problems unless additional constraints are imposed, as illustrated in Example 2.1.

Example 2.1 (Identification failure induced by low-rank predictors)**.** *Consider a simple case* $(p,\, K_0,\, R_0) = (3,\, 2,\, 2)$*:* $\boldsymbol{y}_i = (\boldsymbol{x}_{i,1} + \boldsymbol{x}_{i,2})b_{i1,0} + \boldsymbol{x}_{i,3}b_{i2,0} + \boldsymbol{F}_0\boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i$*, where the third predictor is low-rank,* $\boldsymbol{X}_{,3} = \boldsymbol{g}_{\cdot 1}\boldsymbol{\gamma}_{\cdot 1}^\top$*,* $\boldsymbol{F}_0 = \left(\boldsymbol{g}_{\cdot 2},\quad \boldsymbol{g}_{\cdot 3}\right)$*. Even if the homogeneous-slope identification conditions* $N^{-1}\boldsymbol{L}_0^\top \boldsymbol{P}_{\boldsymbol{\gamma}_{\cdot 1}}^\perp \boldsymbol{L}_0 > 0$ *and* $T^{-1}\boldsymbol{F}_0^\top \boldsymbol{P}_{\boldsymbol{g}_{\cdot 1}}^\perp \boldsymbol{F}_0 > 0$ *hold, the model is not identifiable: an alternative parameterization produces the same fitted values,* $\boldsymbol{y}_i = (\boldsymbol{x}_{i,1} + \boldsymbol{x}_{i,2})b_{i1,0} + \boldsymbol{x}_{i,3}b_{i2,0}^\star + \boldsymbol{F}_0^\star \boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i$*, where* $b_{i2,0}^\star = b_{i2,0} - \left(l_{i1,0} + l_{i2,0}\right)/\gamma_{i1}$ *and* $\boldsymbol{F}_0^\star = \left(\boldsymbol{g}_{\cdot 1} + \boldsymbol{g}_{\cdot 2},\quad \boldsymbol{g}_{\cdot 1} + \boldsymbol{g}_{\cdot 3}\right)$*.*

A common approach in the literature on panels with slope heterogeneity is to remove low-rank predictors entirely. For instance, Avarucci and Zaffaroni (2019) de-mean the data by pre-multiplying each section with $\boldsymbol{P}_{\boldsymbol{G}}^\perp$, leading to the transformed model: $\boldsymbol{y}_i^\star = \boldsymbol{X}_i^\star \boldsymbol{\beta}_{i,0}^{high} + \boldsymbol{F}_0^\star \boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i^\star$, where $\boldsymbol{\beta}_{i,0}^{high}$ are coefficients corresponding to high-rank predictors, and $\boldsymbol{y}_i^\star = \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{y}_i, \boldsymbol{X}_i^\star = \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{X}_i^{high}, \boldsymbol{F}_0^\star = \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{F}_0, \boldsymbol{\varepsilon}_i^\star = \boldsymbol{P}_{\boldsymbol{G}}^\perp \boldsymbol{\varepsilon}_i$. Similarly, Ando and Bai (2015) and Goodhand (2021) implicitly exclude low-rank predictors by assuming $\inf_{\boldsymbol{F}} N^{-1}\sum_{i=1}^N \boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^\top \otimes \left[\boldsymbol{I}_T - \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \left(\boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i\right)^{-1} \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp\right] > 0$. However, in practice, low-rank predictors are often present and may even dominate (see the `French` dataset in Section 4.2). Excluding them could therefore lead to substantial information loss and may not be practical. This motivates our adoption of the constraint $\boldsymbol{F}_0 \perp \boldsymbol{G}$, which allows us to retain low-rank predictors while achieving model identification.

While necessary for identification, this restriction also entails several implications for model structure, which we shall discuss next. When more general low-rank predictors $\boldsymbol{X}_{,j}$ with $\mathrm{Rank}(\boldsymbol{X}_{,j}) = l_j \geq 1$ are present, each $\boldsymbol{X}_{,j}$ must be decomposed into l_j rank-one components for analysis. Otherwise, the model may not be representable in the form of (2.2), as illustrated in Example 2.2. All asymptotics results continue to hold under this generalization, provided that $\{l_j\}_{j=p_1+1}^p$ are finite.

Example 2.2 (The need for rank-one decomposition)**.** *Consider a simple case* $K_0 = p = 1$ *and* $R_0 = 2$*:* $\boldsymbol{y}_i = \boldsymbol{X}_{i,1}b_{i,0} + \boldsymbol{F}_0\boldsymbol{l}_{i,0} + \varepsilon_i$*, where* $\boldsymbol{X}_{,1} = \boldsymbol{g}_{\cdot 1}\boldsymbol{\gamma}_{\cdot 1}^\top + \boldsymbol{g}_{\cdot 2}\boldsymbol{\gamma}_{\cdot 2}^\top$*,* $\boldsymbol{F}_0 = \left(\boldsymbol{g}_{\cdot 2},\quad \boldsymbol{g}_{\cdot 3}\right)$*,* $\boldsymbol{L}_0 = \left(\boldsymbol{\gamma}_{\cdot 2}',\quad \boldsymbol{\gamma}_{\cdot 3}\right)$*, and* $\boldsymbol{g}_{\cdot j_1}^\top \boldsymbol{g}_{\cdot j_2} = 0$ *for* $j_1 \neq j_2$*. This model cannot be written in the form of* (2.2)*. However, if we decompose* $\boldsymbol{X}_{,1}$ *into two rank-one predictors:* $\boldsymbol{X}_{,1}^\star = \boldsymbol{g}_{\cdot 1}\boldsymbol{\gamma}_{\cdot 1}^\top$ *and* $\boldsymbol{X}_{,2}^\star = \boldsymbol{g}_{\cdot 2}\boldsymbol{\gamma}_{\cdot 2}^\top$*, then the model can be expressed as* (2.2)*:*

$$\boldsymbol{y}_i = \boldsymbol{X}_i^\star \boldsymbol{\Theta}_0^\star \boldsymbol{b}_{i,0}^\star + \boldsymbol{F}_0^\star \boldsymbol{l}_{i,0}^\star + \boldsymbol{\varepsilon}_i, \quad s.t.\ \boldsymbol{F}_0^\star \perp \left(\boldsymbol{g}_{\cdot 1},\quad \boldsymbol{g}_{\cdot 2}\right),$$

where $\boldsymbol{X}_i^\star = \left(\boldsymbol{x}_{i,1}^\star,\quad \boldsymbol{x}_{i,2}^\star\right)$*,* $\boldsymbol{\Theta}_0^\star = \boldsymbol{I}_2$*,* $\boldsymbol{b}_{i,0}^\star = \left(b_{i,0},\, b_{i,0} + \gamma_{i2'}/\gamma_{i2}\right)^\top$*,* $\boldsymbol{F}_0^\star = \boldsymbol{g}_{\cdot 3}$*, and* $\boldsymbol{L}_0^\star = \boldsymbol{\gamma}_{\cdot 3}$*, with* $K_0^\star = p^\star = 2$ *and*

$R_0^\star = 1$.

Moreover, imposing the orthogonality constraint can make the resulting model less parsimonious. To satisfy $\boldsymbol{F}_0 \perp \boldsymbol{G}$, the representation may require a larger number of common components K_0 than in the unconstrained case, as demonstrated in Example 2.3. Later in Section 3.5, we select the optimal specification within this orthogonality-constrained class when discussing model selection, i.e., the choice of (K, R).

Example 2.3 (Loss of parsimony under the orthogonality constraint)**.** *Consider a case $K_0 = 1$, $p = 3$, and $R_0 = 2$: $\boldsymbol{y}_i = \boldsymbol{X}_i\boldsymbol{\theta}_{1,0}b_{i,0} + \boldsymbol{F}_0\boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i$, where $\boldsymbol{X}_{\cdot j} = \boldsymbol{g}_{\cdot j}\boldsymbol{\gamma}_{\cdot j}^\top$ for $j = 1, 2, 3$, $\boldsymbol{\theta}_{1,0} = (1, 1, 1)^\top$, $\boldsymbol{F}_0 = \left(\boldsymbol{g}_{\cdot 3} + \boldsymbol{g}_{\cdot 4}, \quad \boldsymbol{g}_{\cdot 5}\right)$, and $\boldsymbol{g}_{\cdot j_1}^\top \boldsymbol{g}_{\cdot j_2} = 0$ for $j_1 \neq j_2$. The model can be expressed as (2.2):*

$$\boldsymbol{y}_i = \boldsymbol{X}_i\boldsymbol{\Theta}_0^\star\boldsymbol{b}_{i,0}^\star + \boldsymbol{F}_0^\star\boldsymbol{l}_{i,0} + \boldsymbol{\varepsilon}_i, \quad s.t.\ \boldsymbol{F}_0^\star \perp \left(\boldsymbol{g}_{\cdot 1}, \quad \boldsymbol{g}_{\cdot 2}, \quad \boldsymbol{g}_{\cdot 3}\right),$$

where $\boldsymbol{\Theta}_0^\star = \left(\boldsymbol{\theta}_{1,0}, \quad \boldsymbol{\theta}_{2,0}\right)$, $\boldsymbol{\theta}_{2,0} = (1, 0, 0)^\top$, $\boldsymbol{b}_{i,0}^\star = (b_{i,0}, l_{i,0})^\top$ and $\boldsymbol{F}_0^\star = \left(\boldsymbol{g}_{\cdot 4}, \quad \boldsymbol{g}_{\cdot 5}\right)$, with $K_0^\star = 2 > K_0$ and $R_0^\star = 2$.

The following assumption will be used to guarantee the model identification.

Assumption 1.

(1) Existence of Second Moments:

$$\mathbb{E}\left(\varepsilon_{it}^2 \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) \leq C_1, \quad \mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^2 \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) \leq C_1, \quad i = 1, \ldots, N, \quad t = 1, \ldots, T.$$

(2) Mean Zero Errors and Exogeneity:

$$\mathbb{E}\left(\varepsilon_{it} \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) = 0, \quad \mathbb{E}\left(\boldsymbol{x}_{it}\varepsilon_{it} \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) = \boldsymbol{0}_p, \quad i = 1, \ldots, N, \quad t = 1, \ldots, T.$$

(3) Condition on High-Rank Predictors if $p_1 > 0$: For any given $\boldsymbol{F} \in \mathbb{R}^{T \times R_0}$,

$$\frac{1}{T}\mathbb{E}\left(\boldsymbol{X}_i^{high\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}^\perp\boldsymbol{X}_i^{high} \middle| \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0\right) \geq C_2, \quad i = 1, \ldots, N,$$

where $\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}^\perp = \boldsymbol{I}_T - \boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}$, $\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}$ is the projection matrix of $\left(\boldsymbol{F}_0, \quad \boldsymbol{F}, \quad \boldsymbol{G}\right)$.

(4) Condition on Low-Rank Predictors if $p_2 > 0$: Conditional on $(\boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0)$,

$$\frac{\boldsymbol{G}^\top\boldsymbol{G}}{T} > 0, \quad \min_{1 \leq i \leq N, p_1+1 \leq j \leq p} |\gamma_{ij}| > 0.$$

Assumption 1(1) and 1(2) follow directly from Moon and Weidner (2017) and are standard. However, the conditions imposed on the high- and low-rank predictors differ from theirs. In Assumption 1(3), the non-collinearity requirement for high-rank predictors is imposed at the individual level, that is, on each

$\boldsymbol{X}_i^{high}$, rather than on the collection $\{\boldsymbol{X}_{,j}\}_{j=1}^p$ across all sections, as in Moon and Weidner (2017). In Assumption 1(4), instead of assuming $T^{-1}\boldsymbol{G}^\top \boldsymbol{P}_{\boldsymbol{F}_0}^\perp \boldsymbol{G} > 0$, we only require $T^{-1}\boldsymbol{G}^\top \boldsymbol{G} > 0$, because the orthogonality constraint $\boldsymbol{F}_0^\top \boldsymbol{G} = \boldsymbol{0}_{R_0 \times p_2}$ is already imposed. In addition, we assume that the elements of $\boldsymbol{\Gamma}$ are uniformly bounded away from zero to guarantee identification of the parameters for each section.

Theorem 2.1 (Identification)**.** *Suppose Assumption 1 holds. Consider the following population-level minimization problem:*

$$\left(\bar{\boldsymbol{\Theta}}, \bar{\boldsymbol{B}}, \bar{\boldsymbol{F}}, \bar{\boldsymbol{L}}\right) = \underset{(\boldsymbol{\Theta}, \boldsymbol{B}, \boldsymbol{F}, \boldsymbol{L}): \boldsymbol{F} \perp \boldsymbol{G}}{\arg\min} \sum_{i=1}^N \mathbb{E}\left(\|\boldsymbol{y}_i - \boldsymbol{X}_i \boldsymbol{\Theta} \boldsymbol{b}_i - \boldsymbol{F} \boldsymbol{l}_i\|_2^2 \,\middle|\, \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0 \right), \tag{2.3}$$

then the solution satisfies $\bar{\boldsymbol{\Theta}}\bar{\boldsymbol{B}}^\top = \boldsymbol{\Theta}_0 \boldsymbol{B}_0^\top$ *and* $\bar{\boldsymbol{F}}\bar{\boldsymbol{L}}^\top = \boldsymbol{F}_0 \boldsymbol{L}_0^\top$.

3 Constrained Principal Component Estimation

In this section, we propose a nonlinear least squares estimator that simultaneously estimates the common component matrix with corresponding loadings $(\boldsymbol{\Theta},\ \boldsymbol{B})$, and latent factors with loadings $(\boldsymbol{F},\ \boldsymbol{L})$ through an iterative optimization procedure.

3.1 Estimation Procedure

Before presenting the estimation algorithm, we impose normalization conditions to ensure the uniqueness of the estimated parameters. These constraints are not related to model-level identification but are necessary to remove the rotational indeterminacy inherent in the factor structure. Specifically, we impose

$$\frac{\boldsymbol{F}^\top \boldsymbol{F}}{T} = \boldsymbol{I}_{R_0}, \quad \frac{\boldsymbol{L}^\top \boldsymbol{L}}{N} = \text{a diagonal matrix of size } R_0, \tag{3.1}$$

and

$$\boldsymbol{\Theta} = \begin{pmatrix} \boldsymbol{I}_{K_0} \\ \boldsymbol{\Theta}_{-[K_0]} \end{pmatrix}. \tag{3.2}$$

We implicitly assume that the first K_0 rows of $\boldsymbol{\Theta}$, $\boldsymbol{\Theta}_{[K_0]}$, are full rank. This assumption entails no loss of generality, since the predictors can always be reordered to satisfy it.

Remark 3.1. *To eliminate rotational indeterminacy, Bai and Ng (2013) discuss three sets of restrictions in detail. The constraints (3.1) adopt the first set of restrictions, and (3.2) adopt the third set.*

Since $\boldsymbol{F}\boldsymbol{L}^\top = \boldsymbol{F}\boldsymbol{O}\boldsymbol{O}^{-1}\boldsymbol{L}^\top$ *for any invertible* $R_0 \times R_0$ *matrix* $\boldsymbol{O}$*, to identify* $(\boldsymbol{F},\ \boldsymbol{L})$ *uniquely from the product* $\boldsymbol{F}\boldsymbol{L}^\top$*, the number of restrictions needed is* R_0^2*. In (3.1), the normalization on* $\boldsymbol{F}$ *provides* $R_0(R_0+1)/2$ *restrictions, and the diagonality of* $\boldsymbol{L}\boldsymbol{L}^\top$ *provides the remaining* $R_0(R_0-1)/2$*. The same scheme is used and discussed in Bai (2009).*

Although similar restrictions could be imposed on $(\boldsymbol{\Theta},\ \boldsymbol{B})$*, such as*

$$\boldsymbol{\Theta}^\top\boldsymbol{\Theta} = \boldsymbol{I}_{K_0}, \quad \frac{\boldsymbol{B}^\top\boldsymbol{B}}{N} = \text{a diagonal matrix of size } K_0,$$

the identification scheme in (3.2) *is more convenient. It fixes the first* $K_0 \times K_0$ *block of* $\boldsymbol{\Theta}$ *to the identity matrix while leaving* $\boldsymbol{B}_0$ *unrestricted. As shown in Section S.2 in the supplementary material, this specification simplifies the derivation of asymptotic results for* $\hat{\boldsymbol{\Theta}}$*, as it avoids enforcing* $\boldsymbol{\Theta}^\top\boldsymbol{\Theta} = \boldsymbol{I}_{K_0}$ *during differentiation.*

We then estimate $(\boldsymbol{\Theta}_0,\ \boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0)$ simultaneously by minimizing the sum of squared residuals:

$$\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{B}}, \hat{\boldsymbol{F}}, \hat{\boldsymbol{L}}\right) = \underset{(\boldsymbol{\Theta},\boldsymbol{B},\boldsymbol{F},\boldsymbol{L}):\boldsymbol{F}\perp\boldsymbol{G}}{\arg\min} \frac{1}{NT}\sum_{i=1}^{N}\|\boldsymbol{y}_i - \boldsymbol{X}_i\boldsymbol{\Theta}\boldsymbol{b}_i - \boldsymbol{F}\boldsymbol{l}_i\|_2^2. \tag{3.3}$$

By solving for $\{\boldsymbol{l}_i\}_{i=1}^N$ in (3.3), we obtain $\{\boldsymbol{l}_i\}_{i=1}^N$ as a function of the parameters $(\boldsymbol{\Theta},\ \boldsymbol{b}_i,\ \boldsymbol{F})$:

$$\boldsymbol{l}_i = \frac{1}{T}\boldsymbol{F}^\top\left(\boldsymbol{y}_i - \boldsymbol{X}_i\boldsymbol{\Theta}\boldsymbol{b}_i\right),$$

and the above minimization is equivalent to

$$\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{B}}, \hat{\boldsymbol{F}}\right) = \underset{(\boldsymbol{\Theta},\boldsymbol{B},\boldsymbol{F}):\boldsymbol{F}\perp\boldsymbol{G}}{\arg\min} \frac{1}{NT}\sum_{i=1}^{N}\|\boldsymbol{P}_{\boldsymbol{F}}^{\perp}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\boldsymbol{\Theta}\boldsymbol{b}_i\right)\|_2^2. \tag{3.4}$$

Given $\hat{\boldsymbol{F}}$, we update $(\boldsymbol{\Theta},\ \boldsymbol{B})$ by iteratively solving least squares problems for $\boldsymbol{\Theta}$ conditional on $\boldsymbol{B}$ and for $\boldsymbol{B}$ conditional on $\boldsymbol{\Theta}$ until convergence. Alternatively, given $\left(\hat{\boldsymbol{\Theta}},\ \hat{\boldsymbol{B}}\right)$, (3.4) implies that $\hat{\boldsymbol{F}}$ solves the constrained principal component problem

$$\frac{1}{NT}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\hat{\boldsymbol{F}} = \hat{\boldsymbol{F}}\boldsymbol{\Lambda}_{NT}, \tag{3.5}$$

where

$$\boldsymbol{\Lambda}_{NT} = \frac{1}{T}\hat{\boldsymbol{F}}^\top\left[\frac{1}{NT}\sum_{i=1}^{N}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)^\top\right]\hat{\boldsymbol{F}} \tag{3.6}$$

is a diagonal matrix containing the largest R_0 eigenvalues of $(NT)^{-1}\sum_{i=1}^{N}\boldsymbol{P}_{\boldsymbol{G}}^{\perp}\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right)^\top\boldsymbol{P}_{\boldsymbol{G}}^{\perp}$, and $\hat{\boldsymbol{F}}$ consists of the corresponding eigenvectors scaled by $\sqrt{T}$. Iterate between updating $\hat{\boldsymbol{F}}$ and $\left(\hat{\boldsymbol{\Theta}},\ \hat{\boldsymbol{B}}\right)$ until convergence. At last, the factor loadings are estimated by

$$\hat{\boldsymbol{l}}_i = \frac{1}{T}\hat{\boldsymbol{F}}^\top\left(\boldsymbol{y}_i - \boldsymbol{X}_i\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{b}}_i\right), \quad i = 1, \ldots, N.$$

Since each block update exactly minimizes the objective function and the criterion is bounded below, the above procedure generates a non-increasing sequence of objective values and converges to a stationary point. This follows from standard results on block coordinate descent; see, e.g., Proposition 2.7.1 in Bertsekas (1999). Given the non-convex nature of the objective function, we employ multiple initializations in practice, which increases the likelihood of converging to the global optimum. A convenient choice of initial values can be

obtained from the nuclear norm regularized estimator (Section S.6); see also Appendix B of Goodhand (2021) for more initialization strategies.

Since $\hat{\boldsymbol{F}}$ is updated by solving the Constrained Principal Component problem (3.5) in an iteration procedure (due to the constraint $\boldsymbol{F} \perp \boldsymbol{G}$), we refer to the resulting procedure as the CPC estimator. In contrast, Bai (2009) refers to his estimator as the PC estimator, because his formulation does not impose the constraint.

3.2 Consistency of Estimators

We make the following assumptions to show the consistency of $\left(\hat{\boldsymbol{\Theta}},\ \hat{\boldsymbol{B}},\ \hat{\boldsymbol{F}}\right)$.

Assumption 2 (Strong Factor Structure)**.**

(1) $T^{-1}\sum_{t=1}^{T} \boldsymbol{f}_{t,0}\boldsymbol{f}_{t,0}^{\top} =: \boldsymbol{\Sigma}_{\boldsymbol{F}} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_{\boldsymbol{F},0}$ *as* $T \to \infty$*, where* $0 < C_3 \le \boldsymbol{\Sigma}_{\boldsymbol{F},0} \le C_4 < \infty$*.*

(2) $N^{-1}\sum_{i=1}^{N} \boldsymbol{l}_{i,0}\boldsymbol{l}_{i,0}^{\top} =: \boldsymbol{\Sigma}_{\boldsymbol{L}} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_{\boldsymbol{L},0}$ *as* $N \to \infty$*, where* $0 < C_3 \le \boldsymbol{\Sigma}_{\boldsymbol{L},0} \le C_4 < \infty$*, and* $\|\boldsymbol{l}_{i,0}\|_2 \le \bar{\boldsymbol{l}}$ *for each* $i = 1, \ldots, N$*.*

(3) $N^{-1}\sum_{i=1}^{N} \boldsymbol{b}_{i,0}\boldsymbol{b}_{i,0}^{\top} := \boldsymbol{\Sigma}_{\boldsymbol{B}} \xrightarrow{\mathbb{P}} \boldsymbol{\Sigma}_{\boldsymbol{B},0}$ *as* $N \to \infty$*, where* $0 < C_3 \le \boldsymbol{\Sigma}_{\boldsymbol{B},0} \le C_4 < \infty$*, and* $\|\boldsymbol{b}_{i,0}\|_2 \le \bar{\boldsymbol{b}}$ *for each* $i = 1, \ldots, N$*.*

Assumption 3 (Weak Exogeneity Condition)**.** *We assume that*

$$\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^{\top}\boldsymbol{\varepsilon}_i\right\|_2^2 = o_p(1).$$

Assumption 4 (Spectral Norm Condition on the Error Matrix)**.** *Let* $\boldsymbol{\mathcal{E}}$ *be a* $T \times N$ *matrix with the* i*-th column* $\boldsymbol{\varepsilon}_i$*. We assume that* $\|\boldsymbol{\mathcal{E}}\|_2 = o_p(\sqrt{NT})$*.*

Assumption 5 (Conditions on Predictors)**.**

(1) Let

$$\mathcal{S}_1 = \left\{\frac{\boldsymbol{X}_i^{\top}\boldsymbol{X}_i}{T} \le C_5, \quad i = 1, \ldots, N\right\}, \quad \mathcal{S}_2 = \left\{\frac{\boldsymbol{X}_i^{\top}\boldsymbol{P}_{\boldsymbol{F}_0}^{\perp}\boldsymbol{X}_i}{T} \ge C_6, \quad i = 1, \ldots, N\right\}.$$

We assume that $\lim_{(N,T)\to\infty}\mathbb{P}(\mathcal{S}_1) = 1$ *and* $\lim_{(N,T)\to\infty}\mathbb{P}(\mathcal{S}_2) = 1$*.*

(2) For the high-rank predictors if $p_1 > 0$*: For any matrix* $\boldsymbol{F} \in \mathbb{R}^{T\times R_0}$*, there exists an index set* $\mathcal{I}_{\boldsymbol{F}} \subset [N]$ *satisfying* $|\mathcal{I}_{\boldsymbol{F}}^c| \le \phi$*, such that for* $i \in \mathcal{I}_{\boldsymbol{F}}$*,*

$$\frac{\boldsymbol{X}_i^{high\top}\boldsymbol{P}_{(\boldsymbol{F}_0,\boldsymbol{F},\boldsymbol{G})}^{\perp}\boldsymbol{X}_i^{high}}{T} \ge C_7. \tag{3.7}$$

Assumptions 2(1) and 2(2) are the standard "strong factor" conditions in panel models with interactive fixed effects; see, e.g., Pesaran (2006), Bai (2009), and Moon and Weidner (2015). We impose the same type of

assumption on $\boldsymbol{B}_0$ in 2(3). While imposing high moment conditions on $\{\boldsymbol{l}_{i,0}\}_{i=1}^N$ and $\{\boldsymbol{b}_{i,0}\}_{i=1}^N$ is more common (e.g., $\mathbb{E}\left(\|\boldsymbol{b}_{i,0}\|_2^8\right)$ and $\mathbb{E}\left(\|\boldsymbol{l}_{i,0}\|_2^8\right)$ are bounded by nonrandom constants uniformly over $i = 1, \ldots, N$), in our setting it would increase the technical complexity of the proofs. We therefore adopt uniform boundedness to simplify the analysis.

Assumption 3 imposes a mild weak exogeneity condition, ensuring that the overall correlation between the predictors and the idiosyncratic errors remains small enough for consistent estimation. Assumption 4, aligned with Assumption 3 in Moon and Weidner (2017), imposes a condition on the spectral norm of the error matrix. This condition is sufficient to ensure the consistency of our estimators. Compared with Assumption 4, Assumption 6(1) further requires that $\|\boldsymbol{\mathcal{E}}\|_2 = O_p\left(M_{NT}\right)$. We discuss this stronger requirement in more details therein.

In Section S.3, We provide some examples for Assumption 5. For Assumption 5(1), the condition $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_1) = 1$ is satisfied in Examples S.3.1 and S.3.2 under the mild growth condition $T^{-1}\log N \to 0$. The condition $\lim_{(N,T)\to\infty} \mathbb{P}(\mathcal{S}_2) = 1$, used to establish the consistency of $\hat{\boldsymbol{B}}$, is verified in Example S.3.4. Together, the events $\mathcal{S}_1$ and $\mathcal{S}_2$ impose uniform upper and lower bounds on the second moments of the predictors. As a sample analogue of Assumption 1(3), since the matrix $\boldsymbol{F}$ is arbitrary, (3.7) in Assumption 5(2) may not hold uniformly for all $i = 1, \ldots, N$. We therefore allow for a finite number of exceptional units and only require that the cardinality of the exceptional set is bounded by a fixed constant ϕ. A concrete specification of ϕ under the condition $T^{-1}\log N \to 0$ is provided in Example S.3.3.

It is worth noting that there is an inherent tradeoff between $|\mathcal{I}_{\boldsymbol{F}}^c|$ and $\|\boldsymbol{\mathcal{E}}\|_2$. To guarantee consistency, it suffices to require $|\mathcal{I}_{\boldsymbol{F}}^c| \cdot \|\boldsymbol{\mathcal{E}}\|_2^2 = o_p(NT)$. If a stronger condition is imposed on the spectral norm of $\boldsymbol{\mathcal{E}}$, for example, $\|\boldsymbol{\mathcal{E}}\|_2 = O_p\left(M_{NT}\right)$ as in Assumption 6(1), then Assumption 5(2) can be correspondingly relaxed to $|\mathcal{I}_{\boldsymbol{F}}^c| = o\left(m_{NT}^2\right)$. This tradeoff is typical: better control of the noise allows for a weaker requirement on the predictors.

Theorem 3.1 (Consistency of Estimators)**.** *Suppose Assumptions 2–5 hold. Then:*

(1) $\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0} - \boldsymbol{\Theta}_0 = o_p(1)$.

(2) The matrix $\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}/T$ *is invertible, and* $\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}} - \boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f = o_p(1)$.

(3) $N^{-1}\left\|\hat{\boldsymbol{B}} - \boldsymbol{B}_0\boldsymbol{\Theta}_{[K_0],0}^\top\right\|_f^2 = o_p(1)$.

3.3 Consistency Rate and Asymptotic Normality

Let $\mathcal{C} = \mathcal{C}_{NT}$ be a sigma field containing the sigma field generated by $(\boldsymbol{B}_0,\ \boldsymbol{F}_0,\ \boldsymbol{L}_0)$. To derive the consistency rate and asymptotic normality of $\hat{\boldsymbol{\Theta}}$ and related bias, we make the following assumptions.

Assumption 6.

(1) The error matrix satisfies $\|\boldsymbol{\mathcal{E}}\|_2 = O_p(M_{NT})$.

(2) $\mathbb{E}\left[\varepsilon_{it} \middle| \mathcal{C} \vee \sigma\{(\boldsymbol{x}_{is}, \varepsilon_{i,s-1}), s \le t\}\right] = 0.$ [1]

(3) $\mathbb{E}\left[\varepsilon_{it}^2 - \sigma_{it}^2 \middle| \mathcal{C} \vee \sigma\{(\boldsymbol{x}_{is}, \varepsilon_{i,s-1}), s \le t\}\right] = 0$, *where* $\sigma_{it}^2 := \mathbb{E}_{\mathcal{C}}\left(\varepsilon_{it}^2\right)$ *for all* $i = 1, \ldots, N$, $t = 1, \ldots, T$.

(4) Conditional on $\mathcal{C}$, $\{(\boldsymbol{x}_{it}, \varepsilon_{it}), t = 1, \ldots, T\}$ *is independent across* i.

(5) $\{(\boldsymbol{x}_{it}, \varepsilon_{it})\}_{t=1}^T$ *is conditional strong mixing given* $\mathcal{C}$ *with the mixing coefficients* $\alpha_i(\cdot)$ *for all* $i = 1, \ldots, N$. *Let* $\alpha(h) = \sup_i \alpha_i(h)$. *We assume that* $\alpha(h) \le h^{-\gamma}$, *where* $\gamma > 2 + 16/\epsilon$ *for some* $\epsilon > 0$.

(6) $\mathbb{E}\left(\varepsilon_{it}^8 \middle| \mathcal{C}\right)$, $\mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^{8+\epsilon} \middle| \mathcal{C}\right)$, $\mathbb{E}\left(\|\boldsymbol{f}_{t,0}\|_2^4\right)$ *are bounded by nonrandom constants uniformly over* $i = 1, \ldots, N$, $t = 1, \ldots, T$.

Compared with Assumption 4, Assumption 6(1) imposes a stronger condition on the spectral norm of the error distribution. See Latala (2005) and Onatski (2015) for related examples and discussions. This assumption is common in the studies of the panel model with IFEs, including Lu and Su (2016), Moon and Weidner (2015), and Armstrong et al. (2024), among others.

Assumption 6(2) assumes that ε_{it} forms a martingale difference sequence (m.d.s.) over time t with respect to the filtration $\mathcal{C} \vee \sigma\{(\boldsymbol{x}_{is}, \varepsilon_{i,s-1}), s \le t\}$, thereby ruling out the serial correlation in the error term. This is standard in the literature; see, e.g., Su and Chen (2013), Moon and Weidner (2017), and Miao et al. (2020). Assumption 6(3), which is less common in dynamic panels with IFEs literature, requires the centered squared error $\varepsilon_{it}^2 - \sigma_{it}^2$ to be a m.d.s. over time. Hence conditional heteroskedasticity is driven by the information captured in $\mathcal{C}$, and is not further predictable by contemporaneous $\boldsymbol{x}_{it}$ or past $(\boldsymbol{x}_{is}, \varepsilon_{is})$, $s \le t-1$, once $\mathcal{C}$ is given. Together 6(2) and 6(3) deliver first- and second-order orthogonality, allowing terms such as $\boldsymbol{\Sigma}_{t=1}^T \boldsymbol{x}_{it}\varepsilon_{it}$ and $\sum_{t=1}^T \boldsymbol{x}_{it}\boldsymbol{x}_{it}^\top\left(\varepsilon_{it}^2 - \sigma_{it}^2\right)$ to be treated as (matrix-valued) martingale sums for concentration and limit arguments.

Assumption 6(4) imposes cross-sectional independence conditional on $\mathcal{C}$, whereas Assumption 6(5) models temporal dependence via a conditional mixing process. Finally, Assumption 6(6) provides uniform higher moment bounds for the errors, predictors, and latent factors, which are used to control tail behavior and justify the application of martingale and mixing inequalities in the asymptotic analysis.

We establish the consistency rate of $\hat{\boldsymbol{F}}$ in the following theorem.

Theorem 3.2 (Consistency Rate of $\hat{\boldsymbol{F}}$)**.** *Suppose Assumptions 2, 5, and 6 hold.*

[1] The notation follows Moon and Weidner (2017), where $\sigma(\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \ldots)$ denotes the sigma field generated by the collection of random variables $\{\boldsymbol{\xi}_1, \boldsymbol{\xi}_2, \ldots\}$, $\mathcal{C}_1 \vee \mathcal{C}_2$ denotes the sigma field generated by the union of all elements in the sigma field $\mathcal{C}_1$ and $\mathcal{C}_2$.

(1) Let $\boldsymbol{O}_{\boldsymbol{F}} = \boldsymbol{\Lambda}_{NT}(\boldsymbol{F}_0^\top \hat{\boldsymbol{F}}/T)^{-1}(\boldsymbol{L}_0^\top \boldsymbol{L}_0/N)^{-1}$*, where* $\boldsymbol{\Lambda}_{NT}$ *is defined in* (3.6)*. Then,*

$$\frac{\|\hat{\boldsymbol{F}}\boldsymbol{O}_{\boldsymbol{F}} - \boldsymbol{F}_0\|_f^2}{T} = O_p\left(\left\|\hat{\boldsymbol{\Theta}}\boldsymbol{\Theta}_{[K_0],0} - \boldsymbol{\Theta}_0\right\|_2^2\right) + O_p\left(\frac{1}{m_{NT}^2}\right).$$

(2) $\boldsymbol{\Lambda}_{NT}$ *is invertible, and* $\boldsymbol{\Lambda}_{NT} \xrightarrow{\mathbb{P}} \boldsymbol{\Lambda}_0$*, where* $\boldsymbol{\Lambda}_0$ *is a diagonal matrix consisting of the eigenvalues of* $\boldsymbol{\Sigma}_{\boldsymbol{F},0}\boldsymbol{\Sigma}_{\boldsymbol{L},0}$*.*

By solving for $\{\boldsymbol{b}_i\}_{i=1}^N$ in (3.4), we obtain $\{\boldsymbol{b}_i\}_{i=1}^N$ as a function of the parameters $(\boldsymbol{\Theta}, \boldsymbol{F})$:

$$\boldsymbol{b}_i = \left(\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\right)^+ \boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{y}_i,$$

and (3.4) is equivalent to

$$\left(\hat{\boldsymbol{\Theta}}, \hat{\boldsymbol{F}}\right) = \underset{(\boldsymbol{\Theta},\boldsymbol{F})}{\arg\min} \frac{1}{NT}\sum_{i=1}^N \left\|\left[\boldsymbol{P}_{\boldsymbol{F}}^\perp - \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\left(\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\right)^+ \boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}}^\perp\right]\boldsymbol{y}_i\right\|_2^2.$$

Given $\hat{\boldsymbol{F}}$, the solution $\hat{\boldsymbol{\Theta}}$ is

$$\hat{\boldsymbol{\Theta}} = \underset{\boldsymbol{\Theta}}{\arg\max} \frac{1}{NT}\sum_{i=1}^N \left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\left(\boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{X}_i \boldsymbol{\Theta}\right)^+ \boldsymbol{\Theta}^\top \boldsymbol{X}_i^\top \boldsymbol{P}_{\hat{\boldsymbol{F}}}^\perp \boldsymbol{y}_i\right\|_2^2. \tag{3.8}$$

We study the asymptotic properties of $\hat{\boldsymbol{\Theta}}$ by exploiting the representation implied by the first-order condition (S.2.10) derived from the objective function (3.8); see Proposition S.2.1. To obtain an explicit asymptotic representation of $\hat{\boldsymbol{\Theta}}$, we introduce the transformed predictors $\{\boldsymbol{Z}_{i,0}\}_{i=1}^N \in \mathbb{R}^{T\times pK_0}$ and define $\boldsymbol{D}(\boldsymbol{F}_0) = (NT)^{-1}\sum_{i=1}^N \boldsymbol{Z}_{i,0}^\top \boldsymbol{Z}_{i,0}$ analogously to Bai (2009); see (S.1.1). Unlike Bai (2009), however, $\boldsymbol{D}(\boldsymbol{F}_0)$ is not strictly positive definite, since $\boldsymbol{D}(\boldsymbol{F}_0)\cdot(\boldsymbol{I}_{K_0}\otimes\boldsymbol{\Theta}_0) = \boldsymbol{0}_{pK_0\times K_0^2}$. To address this degeneracy, we impose positive definiteness on $\boldsymbol{J}\cdot\boldsymbol{D}(\boldsymbol{F}_0)\cdot\boldsymbol{J}^\top$ with

$$\boldsymbol{J} = \boldsymbol{I}_{K_0} \otimes \left(\boldsymbol{0}_{(p-K_0)\times K_0} \quad \boldsymbol{I}_{p-K_0}\right),$$

which corresponds to fixing the first $K_0 \times K_0$ block of $\boldsymbol{\Theta}$ in (3.2). This condition requires sufficient cross-sectional and time-serial variation in $\boldsymbol{J}\cdot\boldsymbol{z}_{it,0}$. We formalize this requirement in the following assumption.

Assumption 7 (Probability Limits of Second Moments)**.** *We assume that the following probability limits exist:*

$$\boldsymbol{D}_0 = \text{plim}_{(N,T)\to\infty}\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T \boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^\top, \quad \boldsymbol{D}_{\boldsymbol{Z}} = \text{plim}_{(N,T)\to\infty}\frac{1}{NT}\sum_{i=1}^N\sum_{t=1}^T \sigma_{it}^2\boldsymbol{z}_{it,0}\boldsymbol{z}_{it,0}^\top,$$

where both $\boldsymbol{J}\boldsymbol{D}_0\boldsymbol{J}^\top$ *and* $\boldsymbol{J}\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{J}^\top$ *are nonrandom and positive definite matrices.*

Assuming $\sqrt{N}/T \to 0$ and $\sqrt{T}/N \to 0$, which are the minimal rate conditions for the asymptotic normality analysis, we show in Section S.2 that $\hat{\boldsymbol{\Theta}}$ admits the following representation:

$$\sqrt{NT}\left[\ell\left(\hat{\boldsymbol{\Theta}}_{-[K]}\boldsymbol{\Theta}_{[K],0}\right) - \ell\left(\boldsymbol{\Theta}_{-[K],0}\right)\right] = \left[\boldsymbol{J}\cdot\boldsymbol{D}\left(\boldsymbol{F}_0\right)\cdot\boldsymbol{J}^\top\right]^{-1}\cdot\frac{1}{\sqrt{NT}}\sum_{i=1}^N \boldsymbol{J}\mathfrak{Z}_{i,0}^\top\boldsymbol{\varepsilon}_i$$

$$+ \boldsymbol{J} \cdot \left[\sqrt{\frac{T}{N}} \mathbb{B}^{\boldsymbol{\xi}} + \sqrt{\frac{N}{T}} \left(\mathbb{B}^{\boldsymbol{\zeta}} + \mathbb{B}^{\boldsymbol{\eta}} \right) \right] + o_p(1), \tag{3.9}$$

where $\left\{ \boldsymbol{\mathfrak{Z}}_{i,0} \right\}_{i=1}^{N}$ are defined in (S.2.14), and $\left(\mathbb{B}^{\boldsymbol{\xi}}, \mathbb{B}^{\boldsymbol{\zeta}}, \mathbb{B}^{\boldsymbol{\eta}} \right)$ are each $O_p(1)$ defined in (S.1.3)–(S.1.5). The terms $\mathbb{B}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\zeta}}$ are the incidental-parameter biases (Neyman and Scott, 1948). The former arises from the imprecise estimation of time effects $\boldsymbol{F}_0$, whereas the latter stems from the estimation error in individual effects $(\boldsymbol{B}_0, \boldsymbol{L}_0)$. Since both sets of nuisance parameters grow with (N, T), they induce non-negligible asymptotic biases. A key step in the analysis is replacing $\boldsymbol{Z}_{i,0}$ with adjusted predictors $\boldsymbol{\mathfrak{Z}}_{i,0}$. The raw sum $\sum_{i=1}^{N} \boldsymbol{Z}_{i,0}^{\top} \boldsymbol{\varepsilon}_i$ contains $\mathbb{B}^{\boldsymbol{\eta}}$, the Nickell bias (Nickell, 1981) arising from the correlation between $\boldsymbol{Z}_{i,0}$ and $\boldsymbol{\varepsilon}_i$. By decomposing this term and explicitly extracting $\mathbb{B}^{\boldsymbol{\eta}}$ (see Corollary S.2.2), the remainder $\sum_{i=1}^{N} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top} \boldsymbol{\varepsilon}_i$ has no bias and satisfies a standard central limit theorem:

$$\frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{J} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top} \boldsymbol{\varepsilon}_i \xrightarrow{\mathbb{D}} \mathcal{N}\left(\boldsymbol{0}_{(p-K_0)K_0}, \boldsymbol{J} \boldsymbol{D}_{\boldsymbol{Z}} \boldsymbol{J}^{\top} \right), \tag{3.10}$$

as shown in Lemma S.2.14. To establish the asymptotic normality of $\hat{\boldsymbol{\Theta}}$, we further make the following assumption.

Assumption 8 (Probability Limits of Bias Components)**.** *We assume that the following probability limits exist:* $\mathbb{B}_0^{\boldsymbol{\xi}} = \operatorname{plim}_{(N,T)\to\infty} \mathbb{B}^{\boldsymbol{\xi}}$, $\mathbb{B}_0^{\boldsymbol{\zeta}} = \operatorname{plim}_{(N,T)\to\infty} \mathbb{B}^{\boldsymbol{\zeta}}$, *and* $\mathbb{B}_0^{\boldsymbol{\eta}} = \operatorname{plim}_{(N,T)\to\infty} \mathbb{B}^{\boldsymbol{\eta}}$.

Theorem 3.3 (Asymptotic Normality of the Estimated Projection - General Case)**.** *Suppose Assumptions 2 and 5–8 hold. If* $T/N \to \rho$, *then,*

$$\sqrt{NT} \left[\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}} \right) - \ell\left(\boldsymbol{P}_{\boldsymbol{\Theta}_0} \right) \right] \xrightarrow{\mathbb{D}} \mathcal{N}\left[\boldsymbol{A}_0^{\top} \left(\rho^{1/2} \mathbb{B}_0^{\boldsymbol{\xi}} + \rho^{-1/2} \mathbb{B}_0^{\boldsymbol{\zeta}} + \rho^{-1/2} \mathbb{B}_0^{\boldsymbol{\eta}} \right), \boldsymbol{A}_0^{\top} \boldsymbol{D}_0^{+} \boldsymbol{D}_{\boldsymbol{Z}} \boldsymbol{D}_0^{+} \boldsymbol{A}_0 \right], \tag{3.11}$$

where $\boldsymbol{A}_0^{\top} \left(\rho^{1/2} \mathbb{B}_0^{\boldsymbol{\xi}} + \rho^{-1/2} \mathbb{B}_0^{\boldsymbol{\zeta}} + \rho^{-1/2} \mathbb{B}_0^{\boldsymbol{\eta}} \right)$ *is the asymptotic bias. Here* $\boldsymbol{A}_0 = \left(\boldsymbol{\Theta}_0^{\top} \boldsymbol{\Theta}_0 \right)^{-1} \boldsymbol{\Theta}_0^{\top} \otimes \boldsymbol{P}_{\boldsymbol{\Theta}_0}^{\perp} \cdot \left(\boldsymbol{I}_{p^2} + \boldsymbol{C}_{p,p} \right)$, *and the commutation matrix* $\boldsymbol{C}_{p,p}$ *transforms* $\ell(\boldsymbol{M})$ *into* $\ell(\boldsymbol{M}^{\top})$ *for any* $\boldsymbol{M} \in \mathbb{R}^{p \times p}$.

Remark 3.2. *Although the normalization matrix* $\boldsymbol{J}$ *depends on which* K_0 *rows of* $\boldsymbol{\Theta}$ *are fixed for identification, the asymptotic distribution in Theorem 3.3 is invariant to this choice. This invariance follows from the fact that the object of interest,* $\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}}$, *depends only on the column space of* $\hat{\boldsymbol{\Theta}}$, *which is unchanged by alternative identifying restrictions.*

When the serial heteroskedasticity is absent, that is, $\sigma_{it}^2 =: \sigma_i^2$ for each $t = 1, \ldots, T$, then $\mathbb{B}^{\boldsymbol{\zeta}} = \boldsymbol{0}_{(p-K_0)K_0}$. In this case, (3.10) can be replaced with

$$\operatorname{plim}_{(N,T)\to\infty} \frac{1}{NT} \sum_{i=1}^{N} \left(\sigma_i^2 \sum_{t=1}^{T} \boldsymbol{z}_{it,0} \boldsymbol{z}_{it,0}^{\top} \right) =: \boldsymbol{D}_1, \quad \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{J} \boldsymbol{\mathfrak{Z}}_{i,0}^{\top} \boldsymbol{\varepsilon}_i \xrightarrow{\mathbb{D}} \mathcal{N}\left(\boldsymbol{0}_{(p-K_0)K_0}, \boldsymbol{J} \boldsymbol{D}_1 \boldsymbol{J}^{\top} \right). \tag{3.12}$$

On the other hand, when the cross-sectional heteroskedasticity is absent, that is, $\sigma_{it}^2 =: \tau_t^2$ for each $i = 1, \ldots, N$, then $\mathbb{B}^{\boldsymbol{\xi}} = \mathbf{0}_{(p-K_0)K_0}$. [2] In this case, (3.10) can be replaced with

$$\operatorname{plim}_{(N,T)\to\infty} \frac{1}{NT} \sum_{t=1}^{T} \left(\tau_t^2 \sum_{i=1}^{N} \boldsymbol{z}_{it,0} \boldsymbol{z}_{it,0}^\top \right) =: \boldsymbol{D}_2, \quad \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{J} \boldsymbol{\mathfrak{Z}}_{i,0}^\top \boldsymbol{\varepsilon}_i \xrightarrow{\mathbb{D}} \mathcal{N}\left(\mathbf{0}_{(p-K_0)K_0}, \ \boldsymbol{J} \boldsymbol{D}_2 \boldsymbol{J}^\top \right). \tag{3.13}$$

Under suitable growth rates of (N, T) and the independence structures mentioned above, all asymptotic bias terms vanish, and the representation (3.9) simplifies to:

$$\sqrt{NT} \left[\ell\left(\hat{\boldsymbol{\Theta}}_{-[K]} \boldsymbol{\Theta}_{0,[K]} \right) - \ell\left(\boldsymbol{\Theta}_{0,-[K]} \right) \right] = \left[\boldsymbol{J} \cdot \boldsymbol{D}\left(\boldsymbol{F}_0 \right) \cdot \boldsymbol{J}^\top \right]^{-1} \cdot \frac{1}{\sqrt{NT}} \sum_{i=1}^{N} \boldsymbol{J} \boldsymbol{\mathfrak{Z}}_{i0}^\top \boldsymbol{\varepsilon}_i + o_p(1).$$

The following result is analogous to that in Bai (2009); see Theorem 2 and Corollary 1 therein.

Theorem 3.4 (Asymptotic Normality of the Estimated Projection - Special Cases)**.** *Suppose Assumptions 2 and 5–8 hold.*

(1) Given $\mathcal{C}$, when the serial heteroskedasticity are absent, and $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$ are conditionally independent of $\{\boldsymbol{X}_i\}_{i=1}^N$, if $\sqrt{N}/T \to 0$ and $T/N \to 0$, then

$$\sqrt{NT} \left[\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}} \right) - \ell\left(\boldsymbol{P}_{\boldsymbol{\Theta}_0} \right) \right] \xrightarrow{\mathbb{D}} \mathcal{N}\left(\mathbf{0}_{p^2}, \ \boldsymbol{A}_0^\top \boldsymbol{D}_0^+ \boldsymbol{D}_1 \boldsymbol{D}_0^+ \boldsymbol{A}_0 \right),$$

where $\boldsymbol{D}_1$ is defined in (3.12).

(2) Given $\mathcal{C}$, when the cross-sectional heteroskedasticity are absent, if $\sqrt{T}/N \to 0$ and $N/T \to 0$, then

$$\sqrt{NT} \left[\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}} \right) - \ell\left(\boldsymbol{P}_{\boldsymbol{\Theta}_0} \right) \right] \xrightarrow{\mathbb{D}} \mathcal{N}\left(\mathbf{0}_{p^2}, \ \boldsymbol{A}_0^\top \boldsymbol{D}_0^+ \boldsymbol{D}_2 \boldsymbol{D}_0^+ \boldsymbol{A}_0 \right),$$

where $\boldsymbol{D}_2$ is defined in (3.13).

(3) Given $\mathcal{C}$, when $\{\varepsilon_{it}\}$ have a constant variance σ^2, and $\{\boldsymbol{\varepsilon}_i\}_{i=1}^N$ are conditionally independent of $\{\boldsymbol{X}_i\}_{i=1}^N$, if $\sqrt{N}/T \to 0$ and $\sqrt{T}/N \to 0$, then

$$\sqrt{NT} \left[\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}} \right) - \ell\left(\boldsymbol{P}_{\boldsymbol{\Theta}_0} \right) \right] \xrightarrow{\mathbb{D}} \mathcal{N}\left(\mathbf{0}_{p^2}, \ \sigma^2 \boldsymbol{A}_0^\top \boldsymbol{D}_0^+ \boldsymbol{A}_0 \right).$$

3.4 Bias Correction and Covariance Matrix Estimation

We provide an analytical bias corrected (ABC) estimator of the asymptotic bias terms $\left(\mathbb{B}^{\boldsymbol{\xi}}, \ \mathbb{B}^{\boldsymbol{\zeta}}, \ \mathbb{B}^{\boldsymbol{\eta}} \right)$ in the representation (3.9) or Theorem 3.3.

To estimate the parameters in Theorem 3.3, we replace $(\boldsymbol{\Theta}_0, \ \boldsymbol{B}_0, \ \boldsymbol{F}_0, \ \boldsymbol{L}_0)$ with $\left(\hat{\boldsymbol{\Theta}}, \ \hat{\boldsymbol{B}}, \ \hat{\boldsymbol{F}}, \ \hat{\boldsymbol{L}} \right)$; see (S.1.6). Let $\hat{\varepsilon}_{it} = y_{it} - \boldsymbol{x}_{it}^\top \hat{\boldsymbol{\Theta}} \hat{\boldsymbol{b}}_i - \hat{\boldsymbol{l}}_i^\top \hat{\boldsymbol{f}}_t$. The diagonal matrix $\boldsymbol{\Omega}_i = \operatorname{Diag}\left(\sigma_{i1}^2, \ \ldots, \ \sigma_{iT}^2 \right)$ is estimated by $\hat{\boldsymbol{\Omega}}_i = \operatorname{Diag}\left(\hat{\varepsilon}_{i1}^2, \ \ldots, \ \hat{\varepsilon}_{iT}^2 \right)$, and $\bar{\sigma}_i^2 = T^{-1} \sum_{i=1}^N \sigma_{it}^2$ is estimated by $\hat{\bar{\sigma}}_i^2 = T^{-1} \sum_{t=1}^T \hat{\varepsilon}_{it}^2$. Then, the bias terms

[2] It is not immediately obvious that the asymptotic bias terms $\mathbb{B}^{\boldsymbol{\zeta}}$ and $\mathbb{B}^{\boldsymbol{\xi}}$ vanish in their respective settings; see the proof of Theorem 3.4 in Section S.2.3 for a detailed explanation.

$\mathbb{B}^{\boldsymbol{\xi}}$ and $\mathbb{B}^{\boldsymbol{\zeta}}$ can be estimated by (S.1.7) and (S.1.8), respectively. In the spirit of Heteroskedasticity and Autocorrelation Consistent estimation (HAC; Newey and West, 1987), Moon and Weidner (2017) estimate $\mathbb{E}_{\mathcal{C}}\left(\boldsymbol{X}_i^\top \boldsymbol{P}_{\boldsymbol{F}_0}\boldsymbol{\varepsilon}_i\right)$ by $T^{-1}\sum_{t=1}^{T}\sum_{s=t+1}^{(t+m)\wedge T}\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^\top\hat{\boldsymbol{f}}_t\hat{\varepsilon}_{it}$ up to a bandwidth m with $m\to\infty$ and $m/T\to 0$, a truncated version of $T^{-1}\sum_{t=1}^{T}\sum_{s=t+1}^{T}\boldsymbol{x}_{is}\hat{\boldsymbol{f}}_s^\top\hat{\boldsymbol{f}}_t\hat{\varepsilon}_{it}$. The same idea applies here to the estimation of $\mathbb{B}^{\boldsymbol{\eta}}$; see (S.1.9). To simplify the analysis of bias correction, we make the following assumption.

Assumption 9. *Both $\|\boldsymbol{f}_{t,0}\|_2$ and $\|\boldsymbol{g}_t\|_2$ are bounded by a constant, uniformly over $t=1,\ldots,T$.*

Theorem 3.5 (Bias Correction for the Estimated Projection)**.** *Suppose Assumptions 2, 5–7, and 9 hold. If $T/N\to\rho$, and the bandwidth m in (S.1.9) satisfies $m\to\infty$ and $m/N^{3/8}\to 0$, then*

$$\sqrt{NT}\left[\ell\left(\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}}\right)-\frac{\hat{\boldsymbol{A}}^\top\hat{\mathbb{B}}^{\boldsymbol{\xi}}}{N}-\frac{\hat{\boldsymbol{A}}^\top(\hat{\mathbb{B}}^{\boldsymbol{\zeta}}+\hat{\mathbb{B}}^{\boldsymbol{\eta}})}{T}-\ell\left(\boldsymbol{P}_{\boldsymbol{\Theta}_0}\right)\right]\xrightarrow{\mathbb{D}}\mathcal{N}\left(\boldsymbol{0}_{p^2},\ \boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0\right). \tag{3.14}$$

For confidence interval construction and hypothesis testing, we also show how to estimate the covariance matrix in the asymptotic distribution of $\tilde{\boldsymbol{\vartheta}}$ consistently. We estimate $\boldsymbol{D}_0$ and $\boldsymbol{D}_{\boldsymbol{Z}}$ by $\hat{\boldsymbol{D}}_0=(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\hat{\boldsymbol{z}}_{it}\hat{\boldsymbol{z}}_{it}^\top$ and $\hat{\boldsymbol{D}}_{\boldsymbol{Z}}=(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\hat{\varepsilon}_{it}^2\hat{\boldsymbol{z}}_{it}\hat{\boldsymbol{z}}_{it}^\top$ respectively, where $\left\{\hat{\boldsymbol{Z}}_i\right\}_{i=1}^{N}$ is defined in (S.1.6).

Proposition 3.1. *Suppose the assumptions in Theorem 3.5 hold. Then, $\hat{\boldsymbol{A}}^\top\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{D}}_{\boldsymbol{Z}}\hat{\boldsymbol{D}}_0^+\hat{\boldsymbol{A}}$ is a consistent estimator of $\boldsymbol{A}_0^\top\boldsymbol{D}_0^+\boldsymbol{D}_{\boldsymbol{Z}}\boldsymbol{D}_0^+\boldsymbol{A}_0$.*

3.5 Model Selection Criteria

In this subsection, we leverage Model Selection Criterion (MSC) for simultaneous selection of the reduced rank parameter K and the number of latent factors R, extending the framework of Bai and Ng (2002) originally proposed for selecting R in pure factor models.

Let $\boldsymbol{\Theta}^{(K)}\in\mathbb{R}^{p\times K}$, $\boldsymbol{B}^{(K)}\in\mathbb{R}^{N\times K}$, $\boldsymbol{F}^{(R)}\in\mathbb{R}^{T\times R}$, and $\boldsymbol{L}^{(R)}\in\mathbb{R}^{N\times R}$ for $K,R\geq 1$. For completeness, we define the case where $K=0$ or $R=0$ by setting

$$\boldsymbol{\Theta}^{(0)}=\boldsymbol{0}_p,\quad \boldsymbol{B}^{(0)}=\boldsymbol{0}_N,\quad \boldsymbol{F}^{(0)}=\boldsymbol{0}_T,\quad \boldsymbol{L}^{(0)}=\boldsymbol{0}_N.$$

Define

$$f_{K,R}^{*}:=\min_{(\boldsymbol{\Theta}^{(K)},\boldsymbol{B}^{(K)},\boldsymbol{F}^{(R)},\boldsymbol{L}^{(R)}):\boldsymbol{F}^{(R)}\perp\boldsymbol{G}}\frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{y}_i-\boldsymbol{X}_i\boldsymbol{\Theta}^{(K)}\boldsymbol{b}_i^{(K)}-\boldsymbol{F}^{(R)}\boldsymbol{l}_i^{(R)}\right\|_2^2 \tag{3.15}$$

for $0\leq K\leq K_{max}$ and $0\leq R\leq R_{max}$, where $K_0<K_{max}\leq p$ and $R_{max}>R_0$. Denote the criterion function as

$$PC(K,R)=f_{K,R}^{*}+K\cdot g_{1,NT}+R\cdot g_{2,NT}, \tag{3.16}$$

where $g_{1,NT}\to 0$, $g_{2,NT}\to 0$, $m_{NT}^2\cdot g_{1,NT}\to\infty$, and $m_{NT}^2\cdot g_{2,NT}\to\infty$. The pair is estimated by

$$\left(\hat{K}_{PC},\hat{R}_{PC}\right)=\underset{0\leq R\leq R_{max},0\leq K\leq K_{max}}{\arg\min}\ PC\left(K,R\right). \tag{3.17}$$

To ensure the model selection consistency, we make the following strengthened assumption.

Assumption 10 (Strengthened Conditions for Model Selection Consistency)**.**

(1) The weak exogeneity condition holds at the rate

$$\frac{1}{NT^2}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^2 = O_p\left(\frac{1}{T}\right).$$

(2) For any matrix $\boldsymbol{F}^{(R)} \in \mathbb{R}^{T\times R}$ with $R \le R_{max}$, there exists an index set $\mathcal{I}_{\boldsymbol{F}^{(R)}} \subset [N]$ satisfying $\left|\mathcal{I}^c_{\boldsymbol{F}^{(R)}}\right| \le \phi$, such that for all $i \in \mathcal{I}_{\boldsymbol{F}^{(R)}}$,

$$\frac{\boldsymbol{X}_i^{high\top}\boldsymbol{P}^{\perp}_{(\boldsymbol{F}_0,\boldsymbol{F}^{(R)},\boldsymbol{G})}\boldsymbol{X}_i^{high}}{T} \ge C_7, \tag{3.18}$$

where $\boldsymbol{G}$ is defined in (2.1).

For model selection consistency, Assumption 10(1) specifies the convergence rate in Assumption 3. This statement is implied by Assumptions 6(2) and 6(6) as

$$\mathbb{E}\left(\frac{1}{NT}\sum_{i=1}^{N}\left\|\boldsymbol{X}_i^\top\boldsymbol{\varepsilon}_i\right\|_2^2\middle|\mathcal{C}\right) = \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^2\cdot\varepsilon_{it}^2\middle|\mathcal{C}\right) \le \frac{1}{NT}\sum_{i=1}^{N}\sum_{t=1}^{T}\left[\mathbb{E}\left(\|\boldsymbol{x}_{it}\|_2^4\middle|\mathcal{C}\right)\cdot\mathbb{E}\left(\varepsilon_{it}^4\middle|\mathcal{C}\right)\right]^{1/2}.$$

Assumption 10(2) strengthens condition (3.7) in Assumption 5(2). The key difference is that the projection space is enlarged to $\left(\boldsymbol{F}_0,\quad \boldsymbol{F}^{(R)},\quad \boldsymbol{G}\right)$ with $R_0 < R \le R_{max}$. This extension is required for consistent model selection when considering overfitted models with $R > R_0$.

Theorem 3.6. *Suppose Assumptions 2, 5(1), 6(1) and 10 hold. Then, (K_0, R_0) can be consistently estimated by (3.17), that is,*

$$\lim_{(N,T)\to\infty}\mathbb{P}\left[\left(\hat{K}_{PC},\hat{R}_{PC}\right) = (K_0,R_0)\right] = 1.$$

Specific choices of PC in (3.16) are as follows:

$$\begin{aligned}
PC_{p1}(K,R) =& f^*_{K,R} + \hat{\sigma}^2\left(c_1\cdot K + c_2\cdot R\right)\left(\frac{N+T}{NT}\right)\ln\left(\frac{NT}{N+T}\right),\\
PC_{p2}(K,R) =& f^*_{K,R} + \hat{\sigma}^2\left(c_1\cdot K + c_2\cdot R\right)\left(\frac{N+T}{NT}\right)\ln m^2_{NT},\\
PC_{p3}(K,R) =& f^*_{K,R} + \hat{\sigma}^2\left(c_1\cdot K + c_2\cdot R\right)\left(\frac{\ln m^2_{NT}}{m^2_{NT}}\right),
\end{aligned} \tag{3.19}$$

where $f^*_{K,R}$ is defined in (3.15), and c_1 and c_2 are prespecified fixed constants serving as penalty weights. The error variance estimator $\hat{\sigma}^2$ for $(NT)^{-1}\sum_{i=1}^{N}\sum_{t=1}^{T}\mathbb{E}\left(\varepsilon_{it}^2\right)$ is given by

$$\hat{\sigma}^2 = \frac{NT}{NT - K_{max}(N+p-K_{max}) - R_{max}(N+T-R_{max})}f^*_{K_{max},R_{max}}.$$

The statistics in (3.19) are referred to as the Panel C_p (PC_p) criteria, as they extend Mallow's C_p (Mallows, 1973) to the panel data setting.

Define

$$IC(K,R) = \ln\left(f^*_{K,R}\right) + K \cdot g_{1,NT} + R \cdot g_{2,NT}.$$

In a manner analogous to Corollary 1 in Bai and Ng (2002), the pair (K_0, R_0) can also be consistently estimated by

$$\left(\hat{K}_{IC}, \hat{R}_{IC}\right) = \underset{0 \le R \le R_{max}, 0 \le K \le K_{max}}{\arg\min} IC(K,R). \tag{3.20}$$

Accordingly, we may select (K, R) which minimizes one of the following functions,

$$\begin{aligned}
IC_{p1}(K,R) &= \ln\left(f^*_{K,R}\right) + (c_1 \cdot K + c_2 \cdot R)\left(\frac{N+T}{NT}\right)\ln\left(\frac{NT}{N+T}\right),\\
IC_{p2}(K,R) &= \ln\left(f^*_{K,R}\right) + (c_1 \cdot K + c_2 \cdot R)\left(\frac{N+T}{NT}\right)\ln m^2_{NT},\\
IC_{p3}(K,R) &= \ln\left(f^*_{K,R}\right) + (c_1 \cdot K + c_2 \cdot R)\left(\frac{\ln m^2_{NT}}{m^2_{NT}}\right),
\end{aligned} \tag{3.21}$$

where the logarithmic transformation in IC_p plays a role analogous to the scale adjustment by $\hat{\sigma}^2$ in PC_p. In our empirical evaluation, we set $c_1 = 1$ and $c_2 = 2$ for both PC_p and IC_p criteria.

4 Empirical Evaluation

We examine the finite-sample behavior of the proposed MSC for determining the structural dimensions (K, R), and assess the associated estimation performance under both simulated and real data settings.

4.1 Simulation Study

We conduct a simulation study to evaluate both model selection and estimation performance of RRPM-IFEs under controlled data-generating processes.

Model Design. The data are generated according to the following model:

$$y_{it} = \boldsymbol{x}_{it}^\top \boldsymbol{\Theta}_0 \boldsymbol{b}_{i,0} + \boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0} + \sigma \varepsilon_{it}, \tag{4.1}$$

where $\boldsymbol{\Theta}_0 \in \mathbb{R}^{p \times K_0}$ is a reduced-rank coefficient matrix with orthogonal columns, $\boldsymbol{b}_{i,0} \in \mathbb{R}^{K_0}$ is the loading vector for individual i, $\boldsymbol{f}_{t,0} \in \mathbb{R}^{R_0}$ is the latent factor vector at time t, $\boldsymbol{l}_{i,0} \in \mathbb{R}^{R_0}$ is the corresponding factor loading vector, and $\sigma > 0$ controls the signal-to-noise ratio.

For the latent factor structure $\boldsymbol{l}_{i,0}^\top \boldsymbol{f}_{t,0}$, the loadings $\{\boldsymbol{l}_{i,0}\}_{i=1}^N$ are drawn independently from $\text{Unif}\left[-\frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2}\right]^{R_0}$, and the factor process $\{\boldsymbol{f}_{t,0}\}_{t=1}^T$ follows $\boldsymbol{f}_{t,0} = \rho_{\boldsymbol{f}} \boldsymbol{f}_{t-1,0} + \sqrt{1-\rho_{\boldsymbol{f}}^2}\boldsymbol{u}_{1t}$, where $\rho_{\boldsymbol{f}} = 0.5$ and $\{\boldsymbol{u}_{1t}\}_{t=1}^T$ are i.i.d. draws from $\mathcal{N}(\boldsymbol{0}_{R_0}, \boldsymbol{I}_{R_0})$. The first $(p-2)$ predictors $\boldsymbol{x}_{it,1:(p-2)}$ are generated independently across i and t from $\mathcal{N}(\boldsymbol{0}_{p-2}, \boldsymbol{\Sigma}_{\boldsymbol{x}})$, with $(\boldsymbol{\Sigma}_{\boldsymbol{x}})_{jk} = 0.6^{|j-k|}$, $j, k = 1, \ldots, p-2$, and $p = 10$. The $(p-1)$-th predictor is low-rank

and given by $\boldsymbol{X}_{,p-1} = \boldsymbol{g}\boldsymbol{\gamma}^{\top}$, where $\boldsymbol{g} = \boldsymbol{P}^{\perp}_{\boldsymbol{F}_0}\boldsymbol{u}_2$, $\boldsymbol{u}_2 \sim \mathcal{N}(\boldsymbol{0}_T,\ \boldsymbol{\Sigma}_{\boldsymbol{u}_2})$ with $(\boldsymbol{\Sigma}_{\boldsymbol{u}_2})_{st} = 0.6^{|s-t|}$, $s, t = 1, \ldots, T$, and $\boldsymbol{\gamma} \sim \mathrm{Unif}[0.5,\ 1.5]^N$. The last predictor $x_{it,p} = y_{i,t-1}$ is the one-period lag of the response and captures time dynamics. The coefficient matrix $\boldsymbol{\Theta}_0$ takes the form

$$\begin{pmatrix} \boldsymbol{\theta}_{1,1:(p-1),0} & \boldsymbol{\theta}_{2,1:(p-1),0} & \cdots & \boldsymbol{\theta}_{K_0,1:(p-1),0} \\ \rho_y & 0 & \cdots & 0 \end{pmatrix},$$

where the first $(p-1)$ rows are constructed as the orthonormal left singular vector matrix of a random Gaussian matrix, and $\rho_y = 0.5$. The loading vectors $\boldsymbol{b}_{i,0}$ are drawn independently from $\mathrm{Unif}[0.5, 1.5]^{K_0}$. The noise level is set as $\sigma = 0.5\sqrt{\max(R_0, 1)}$.

Simulation Settings. We consider the following combinations of sample sizes: $(N, T) \in \{(20, 200),\ (50, 200), (100, 200), (200, 200), (200, 20), (200, 50), (200, 100)\}$. These choices allow us to examine different relative magnitudes of the cross-sectional and time dimensions. We also consider the following configurations of structural dimensions: $(K_0, R_0) \in \{(2, 2), (0, 2), (2, 0), (0, 0)\}$, which correspond to a full model, a pure factor model, a pure reduced-rank regression model, and a null model, respectively. To assess robustness, the idiosyncratic error ε_{it} is generated under three designs: i.i.d. normal (E1) with $\varepsilon_{it} \sim \mathcal{N}(0, 1)$, heavy-tailed (E2) with $\varepsilon_{it} \sim t_{10}$, a Student's t-distribution with degree of freedom 10, and heteroskedastic (E3) where the variance changes with section and time: $\sigma^2_{it} = 2/3$ if $i + t$ is odd, and $\sigma^2_{it} = 4/3$ if $i + t$ is even.

Estimation, Model Selection, and Evaluation. For each simulated dataset, parameter estimation and model selection proceed as follows. We first estimate $\left(\hat{\boldsymbol{\Theta}}^{(K)},\ \hat{\boldsymbol{B}}^{(K)},\ \hat{\boldsymbol{F}}^{(R)},\ \hat{\boldsymbol{L}}^{(R)}\right)$ by solving (3.3) or (3.4) using an iterative procedure. The pair (K, R) is then selected using the model selection criteria defined in (3.17) and (3.20).

Estimation accuracy is evaluated using Frobenius norms of projection matrix differences, $\left\|\boldsymbol{P}_{\hat{\boldsymbol{\Theta}}} - \boldsymbol{P}_{\boldsymbol{\Theta}_0}\right\|_f$ and $\left\|\boldsymbol{P}_{\hat{\boldsymbol{F}}} - \boldsymbol{P}_{\boldsymbol{F}_0}\right\|_f$, where the true dimensions (K_0, R_0) are assumed known. For each configuration, 100 replications are conducted. The average of the estimation accuracy is calculated, and we also record the empirical frequencies with which each candidate (K, R) is selected, highlighting the most frequently chosen values.

Simulation Results. To save space, only the results of PC_{p1} and IC_{p1} in MSC are reported here. Full simulation tables including PC_{p2}–PC_{p3} and IC_{p2}–IC_{p3} are provided in Section S.2. We summarize the key findings as follows.

Table 1 reports results for the full model $(K_0, R_0) = (2, 2)$ under three error designs, i.i.d. normal (E1), heavy-tailed (E2), and heteroskedastic (E3), as specified in the simulation settings. Across all designs, the estimation errors for $\boldsymbol{\Theta}$ and $\boldsymbol{F}$ decrease steadily as the sample size grows. Both PC_{p1} and IC_{p1} identify the

Table 1: Simulation results under three error types under $(K_0, R_0) = (2, 2)$.

Error type	(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	**K=2**	K=3	K=4	R=0	R=1	**R=2**	R=3	R=4
E1 (i.i.d. normal)	(20, 200)	0.2124	0.6365	PCp1	0	34	**66**	0	0	0	10	**90**	0	0
				ICp1	0	43	**57**	0	0	4	24	**72**	0	0
	(50, 200)	0.1342	0.4048	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(100, 200)	0.0921	0.2863	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 200)	0.0617	0.2019	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 20)	0.2461	0.2270	PCp1	0	4	**96**	0	0	0	29	**71**	0	0
				ICp1	0	7	**93**	0	0	4	33	**63**	0	0
	(200, 50)	0.1351	0.2100	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 100)	0.0921	0.2070	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
E2 (heavy-tailed)	(20, 200)	0.2126	0.6328	PCp1	0	42	**58**	0	0	0	6	**94**	0	0
				ICp1	0	47	**53**	0	0	6	17	**77**	0	0
	(50, 200)	0.1219	0.4037	PCp1	0	1	**99**	0	0	0	0	**100**	0	0
				ICp1	0	2	**98**	0	0	0	0	**100**	0	0
	(100, 200)	0.0888	0.2847	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 200)	0.0632	0.2016	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 20)	0.2508	0.2399	PCp1	0	6	**94**	0	0	2	31	**67**	0	0
				ICp1	0	14	**86**	0	0	12	34	**54**	0	0
	(200, 50)	0.1325	0.2120	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 100)	0.0865	0.2022	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
E3 (heteroskedastic)	(20, 200)	0.2125	0.6366	PCp1	0	34	**66**	0	0	0	10	**90**	0	0
				ICp1	0	41	**59**	0	0	5	23	**72**	0	0
	(50, 200)	0.1333	0.4047	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	1	**99**	0	0	0	0	**100**	0	0
	(100, 200)	0.0923	0.2867	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 200)	0.0619	0.2015	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 20)	0.2508	0.2262	PCp1	0	3	**97**	0	0	0	32	**68**	0	0
				ICp1	0	8	**92**	0	0	4	33	**63**	0	0
	(200, 50)	0.1362	0.2105	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0
	(200, 100)	0.0920	0.2072	PCp1	0	0	**100**	0	0	0	0	**100**	0	0
				ICp1	0	0	**100**	0	0	0	0	**100**	0	0

correct dimensions accurately once the sample size is moderate, while some under-selection may occur in small samples such as $(20, 200)$ or $(200, 20)$. Overall, the results are very similar across E1–E3, indicating that the proposed criteria remain stable under non-Gaussian and heteroskedastic errors. Table 2 fixes the error design at E1 and considers different structural specifications. For $(K_0, R_0) = (0, 2)$, the criteria always select $K = 0$ and identify $R = 2$ with high frequency. For $(K_0, R_0) = (2, 0)$, the correct dimensions $K = 2$ and $R = 0$ are selected in nearly all cases. In the null model $(K_0, R_0) = (0, 0)$, the criteria always select $K = 0$ and $R = 0$, indicating that no spurious structures are introduced.

Taken together, the simulation results show that the proposed procedures achieve reliable model selection across different error designs and structural configurations, with estimation accuracy improving as the sample size increases.

Table 2: Simulation results under a fixed error setting E_1 for different (K_0, R_0) configurations.

(K_0, R_0)	(N, T)	Err.$\boldsymbol{\Theta}$	Err.$\boldsymbol{F}$	Method	K=0	K=1	K=2	K=3	K=4	R=0	R=1	R=2	R=3	R=4
(0, 2)	(20, 200)	–	0.6333	PCp1	**100**	0	0	0	0	0	9	**91**	0	0
				ICp1	**100**	0	0	0	0	1	18	**81**	0	0
	(50, 200)	–	0.4027	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
				ICp1	**100**	0	0	0	0	0	0	**100**	0	0
	(100, 200)	–	0.2854	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
				ICp1	**100**	0	0	0	0	0	0	**100**	0	0
	(200, 200)	–	0.2010	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
				ICp1	**100**	0	0	0	0	0	0	**100**	0	0
	(200, 20)	–	0.2083	PCp1	**100**	0	0	0	0	0	18	**82**	0	0
				ICp1	**100**	0	0	0	0	5	33	**62**	0	0
	(200, 50)	–	0.2059	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
				ICp1	**100**	0	0	0	0	0	0	**100**	0	0
	(200, 100)	–	0.2055	PCp1	**100**	0	0	0	0	0	0	**100**	0	0
				ICp1	**100**	0	0	0	0	0	0	**100**	0	0
(2, 0)	(20, 200)	0.1459	–	PCp1	0	1	**99**	0	0	**100**	0	0	0	0
				ICp1	0	3	**97**	0	0	**100**	0	0	0	0
	(50, 200)	0.0885	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
	(100, 200)	0.0630	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
	(200, 200)	0.0445	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
	(200, 20)	0.1563	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
	(200, 50)	0.0892	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
	(200, 100)	0.0612	–	PCp1	0	0	**100**	0	0	**100**	0	0	0	0
				ICp1	0	0	**100**	0	0	**100**	0	0	0	0
(0, 0)	(20, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(50, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(100, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(200, 200)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(200, 20)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(200, 50)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0
	(200, 100)	–	–	PCp1	**100**	0	0	0	0	**100**	0	0	0	0
				ICp1	**100**	0	0	0	0	**100**	0	0	0	0

4.2 Real Data Analysis

We apply our proposed method to two real-world datasets to illustrate its usefulness in data analysis.

Ken French's Dataset: We use the 100 portfolios data with 5 factors. The dataset consists of returns on $N = 100$ value-weighted portfolios and $T = 1258$ daily observations spanning January 2, 2019 to December 29, 2023. The response variable is the portfolio return. The predictors are the $p = 5$ Fama–French factors (Fama and French, 1993, 2015): `MKT − RF` (market excess return), `SMB` (size), `HML` (value), `RMW` (profitability), and `CMA` (investment). All factors are time-varying but cross-sectionally invariant, and therefore enter the model as low-rank predictors. Detailed descriptions of the dataset are available at https://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html.

Table 3 summarizes the selected (K, R) based on model selection criteria IC_{p1}–IC_{p3} and PC_{p1}–PC_{p3}. For the `French` dataset, all criteria agree on the reduced-rank dimension $K = 4$. However, IC_{p1}–IC_{p3} select $R = 0$, showing that there is no latent factor structure, while PC_{p1}–PC_{p3} select $R = 1$, indicating the potential existence of a latent factor.

Table 3: Selected (K, R) combinations for two real datasets using MSC.

Data	ICp1	ICp2	ICp3	PCp1	PCp2	PCp3
`French`	(4, 0)	(4, 0)	(4, 0)	(4, 1)	(4, 1)	(4, 1)
`PWT`	(2, 1)	(2, 1)	(2, 1)	(4, 3)	(4, 3)	(6, 6)

Table 4: Explanatory power and singular value decomposition under $(K, R) = (4, 1)$ for `French`.

Fama–French Factors	$\boldsymbol{\theta}_1$	$\boldsymbol{\theta}_2$	$\boldsymbol{\theta}_3$	$\boldsymbol{\theta}_4$	$\boldsymbol{f}_{\cdot 1}$	only $\boldsymbol{X}$
`MKT − RF`	**-0.931**	0.346	0.081	-0.062	–	–
`SMB`	-0.342	**-0.928**	-0.074	-0.013	–	–
`HML`	-0.119	0.018	**-0.696**	0.458	–	–
`RMW`	0.048	0.139	-0.624	-0.037	–	–
`CMA`	-0.007	0.007	0.338	**0.886**	–	–
Cumulative R^2	0.7955	0.8221	0.8503	0.8628	0.8744	0.8641

Table 4 compares the coefficients of determination (R^2) across different model specifications. Using four linear combinations of the five factors explains $R^2 = 0.8628$ of the total variation. In comparison, using all five factors directly yields only a marginal increase in explanatory power, with the last direction contributing merely 0.0013 additional variation. This finding provides empirical evidence for the presence of a reduced-rank structure in this dataset. Table 4 also reports the singular value decomposition (SVD) of the estimated matrix $\hat{\boldsymbol{\Theta}}\hat{\boldsymbol{B}}^\top$. The corresponding singular values are 14.565, 3.075, 2.743, and 2.023. The first direction is dominated by the market excess return, with an absolute loading of 0.931 on `MKT − RF` and a substantially larger singular value than the remaining directions, indicating that aggregate market risk dominates the linear-combination structure. The second direction is mainly associated with the size factor `SMB`. The third direction reflects a joint contribution of the value factor `HML` and the profitability factor `RMW`. The fourth direction is primarily driven by the investment factor `CMA`.

Regarding the latent factor dimension, the selection of $R = 1$ by PC_{p1}–PC_{p3} indicates that a purely covariate-driven specification may be insufficient to fully explain the return dynamics. Incorporating a latent factor increases the cumulative R^2 from 0.8628 to 0.8744, an improvement of 0.0116, which exceeds the gain 0.0013 obtained from using all five factors directly. This highlights the importance of interactive fixed effects in capturing residual cross-sectional dependence in the `French` dataset.

To interpret the latent factor, we divide the 100 portfolios into 10 groups according to size, where 1

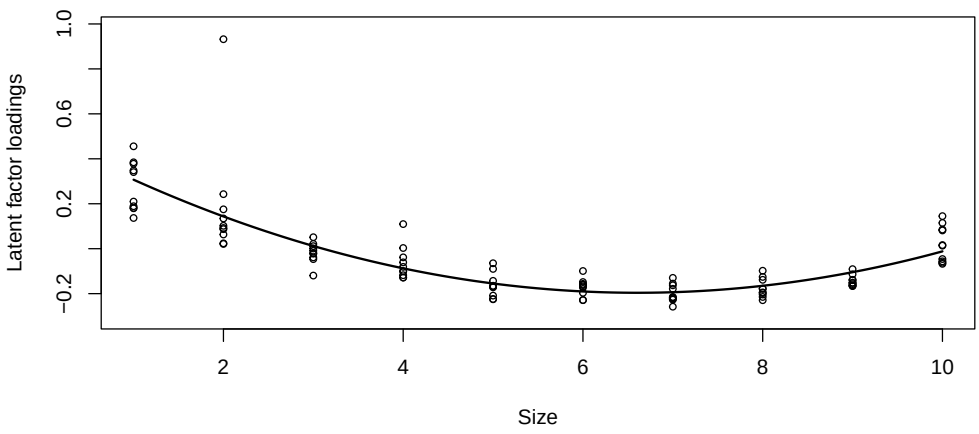

Figure 1: Latent factor loadings against size with a fitted quadratic curve.

denotes the smallest and 10 the largest. As shown in Figure 1, the estimated loadings $\hat{\boldsymbol{l}}_{\cdot 1}$ display a clear quadratic pattern with respect to size. The fitted curve $\hat{y} = 0.503 - 0.211x + 0.016x^2$ achieves an R^2 of 0.696. Motivated by this pattern, we construct a proxy for size curvature as (weighted return of size 1 + weighted return of size 10)/2 − weighted return of size 5 and 6. This proxy has a correlation of 0.554 with the estimated latent factor $\hat{\boldsymbol{f}}_{\cdot 1}$. This suggests that the latent factor can be interpreted as a contrast across market capitalization, defined as

$$(\text{Big-cap return} + \text{Small-cap return})/2 - \text{Medium-cap return},$$

which complements the size factor `SMB`. Economically, this size curvature reflects heterogeneous investor demand across market capitalization: speculative investors tend to pursue high-α opportunities in small-cap stocks, while more conservative investors tilt toward large-cap stocks, which are generally more stable and may benefit from policy support. Medium-cap stocks, by contrast, combine features of both segments, balancing growth potential and stability. This intermediate positioning naturally gives rise to the observed curvature pattern. In conclusion, the latent factor captures a nonlinear size effect characterized by a curvature across market capitalization, rather than a simple monotonic size premium. This finding aligns with the "mid-cap effect" documented in the literature; see, e.g., Menchero et al. (2008) and Ruban et al. (2012).

Penn World Table (PWT) 10.01 We use a balanced panel constructed from the `Penn World Table` (version 10.01), consisting of $N = 91$ countries observed over $T = 50$ annual periods spanning 1970–2019. The response variable is real GDP per capita, measured as log(`rgdpe/pop`). The predictor set contains $p = 11$ macroeconomic covariates, including both contemporaneous values and selected lagged terms. Specifically, the predictors include lag 1 and lag 2 of log(`rgdpe/pop`), the contemporaneous and lag 1 of log(`rnna/emp`) (capital stock per worker), log(`hc`) (human capital index), log(`emp/pop`) (employment-to-population ratio),

log(pl_i/pl_c) (relative investment price) together with its one-period lag, as well as csh_i (investment share), csh_g (government consumption share), and a rank-one time trend variable year. The data are available at https://www.rug.nl/ggdc/productivity/pwt/.

For the PWT dataset, the IC criteria agree on the parsimonious choice $(K, R) = (2, 1)$, whereas the PC criteria select substantially larger K and R. This discrepancy may reflect the difference in objective scales: $\log\left(\hat{f}_{K,R}\right)$ compresses differences in fit under IC, while PC compares $\hat{f}_{K,R}$ on the original scale and scales the complexity penalty by $\hat{\sigma}^2$. Consequently, changes in residual scale and fit magnitude can have a larger effect on the effective fit–complexity trade-off under PC, leading to greater variability in the selected (K, R) in only moderately large samples.

Table 5: Explanatory power and singular value decomposition under $(K, R) = (2, 1)$ for PWT.

Predictors	$\boldsymbol{\theta}_1$	$\boldsymbol{\theta}_2$	$\boldsymbol{f}_{\cdot 1}$	only $\boldsymbol{X}$
log(rgdpe/pop)$_{t-1}$	**-0.989**	0.003	–	–
log(rgdpe/pop)$_{t-2}$	0.143	-0.003	–	–
log(rnna/emp)$_t$	-0.019	-0.479	–	–
log(rnna/emp)$_{t-1}$	-0.025	**0.637**	–	–
log(hc)	-0.024	-0.519	–	–
log(emp/pop)	-0.016	0.067	–	–
log(pl_i/pl_c)$_t$	-0.003	0.027	–	–
log(pl_i/pl_c)$_{t-1}$	-0.001	0.004	–	–
csh_i	-0.010	0.047	–	–
csh_g	0.008	0.018	–	–
year	-0.002	0.296	–	–
Cumulative R^2	0.9966	0.9970	0.9980	0.9984

Table 5 shows that reducing the predictor dimension from 11 variables to two linear combinations entails some loss in fit, with the cumulative R^2 decreasing by 0.0014. Introducing a single latent factor recovers 0.0010 of the explained variation, yielding a cumulative R^2 close to that obtained when all predictors are included directly. Overall, the joint use of reduced-rank predictors and latent factors achieves substantial dimension reduction with only a minor deterioration in model fit. Additional results reported in Section S.3 further show that, under alternative specifications such as $(K, R) = (4, 3)$, the proposed approach can even outperform the specifications using all predictors directly in terms of explanatory power.

Table 5 also reports the top contributors to the two leading directions under $(K, R) = (2, 1)$. The first direction is dominated by the first two lags of real GDP per capita, indicating strong persistence and dynamic dependence in economic growth. The second direction loads primarily on capital intensity and human capital, with substantial contributions from log(rnna/emp) and log(hc), as well as a moderate contribution from the time trend. This direction reflects a structural development component associated with capital deepening

and long-run growth patterns. We further find that the key predictors remain important under alternative specifications selected by PC_{p1}–PC_{p3}; see Section S.3 for other SVD results.

The latent factor in this data set is partially associated with observable macroeconomic variables, including total factor productivity (`ctfp`), average working hours (`avh`), and labor compensation share (`labsh`), which are excluded from the model due to limited availability in earlier periods. We compute country-level correlations between the factor and `ctfp` for countries with at least 30 observations, finding an average correlation of about 0.24. The associations with `avh` and `labsh` are weaker but still often statistically significant. These results suggest that the latent factor captures some omitted but economically relevant variation. However, the findings should be interpreted with caution due to potential measurement error and lagged effects in macroeconomic panel data; see Durlauf et al. (2005).

5 Conclusion and Extensions

This paper develops a reduced-rank panel model with interactive fixed effects to capture structured slope heterogeneity under cross-sectional dependence, while retaining parsimony and interpretability. Identification of the model is established, and estimation is carried out via a Constrained Principal Component approach. Under large panel asymptotics with both $(N, T) \to \infty$, we derive the asymptotic properties of the proposed estimator and provide data-driven procedures for selecting the structural dimensions of slope heterogeneity and latent factors. Numerical evidence confirms the usefulness of the proposed selection criteria and illustrates the empirical interpretability of the recovered latent slope patterns.

Several directions for future work are suggested by the present analysis. In fully heterogeneous models, the orthogonality constraint is essential when predictors exhibit low-rank structures, as slope parameters may be confounded with latent factors, leading to identification failure. In the reduced-rank model, such constraint is mainly imposed to rule out pathological cases (see Example 2.1). A simple sufficient condition under which the constraint can be omitted is $\text{Rank}\left(\mathbf{\Theta}_0^{high}\right) = K_0$, where $\mathbf{\Theta}_0^{high}$ denotes the slope coefficients associated with high-rank predictors. This condition fails when the model contains only low-rank predictors (e.g., `French` dataset). Developing more general identification conditions for reduced-rank models, akin to those in Moon and Weidner (2017) for the homogeneous slope case, remains an intriguing direction for future research.

The current model assumes that slope heterogeneity is entirely driven by a common reduced-rank structure. In practice, however, additional individual-specific deviations may arise. A natural extension is to augment the reduced-rank specification with an individual-specific component, allowing the slopes to be decomposed into a low-dimensional structured component that captures systematic heterogeneity, and a

residual idiosyncratic term. This formulation retains dimension reduction while accommodating more flexible cross-sectional heterogeneity.

Supplementary Material: Supplementary materials, including the R code used in the analysis, are available at https://github.com/zmk-98/RRPM-IFEs.

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